SURFACE ANATOMY OF THE UPPER LIMB: Practical Workshop

PT7001/PT4054

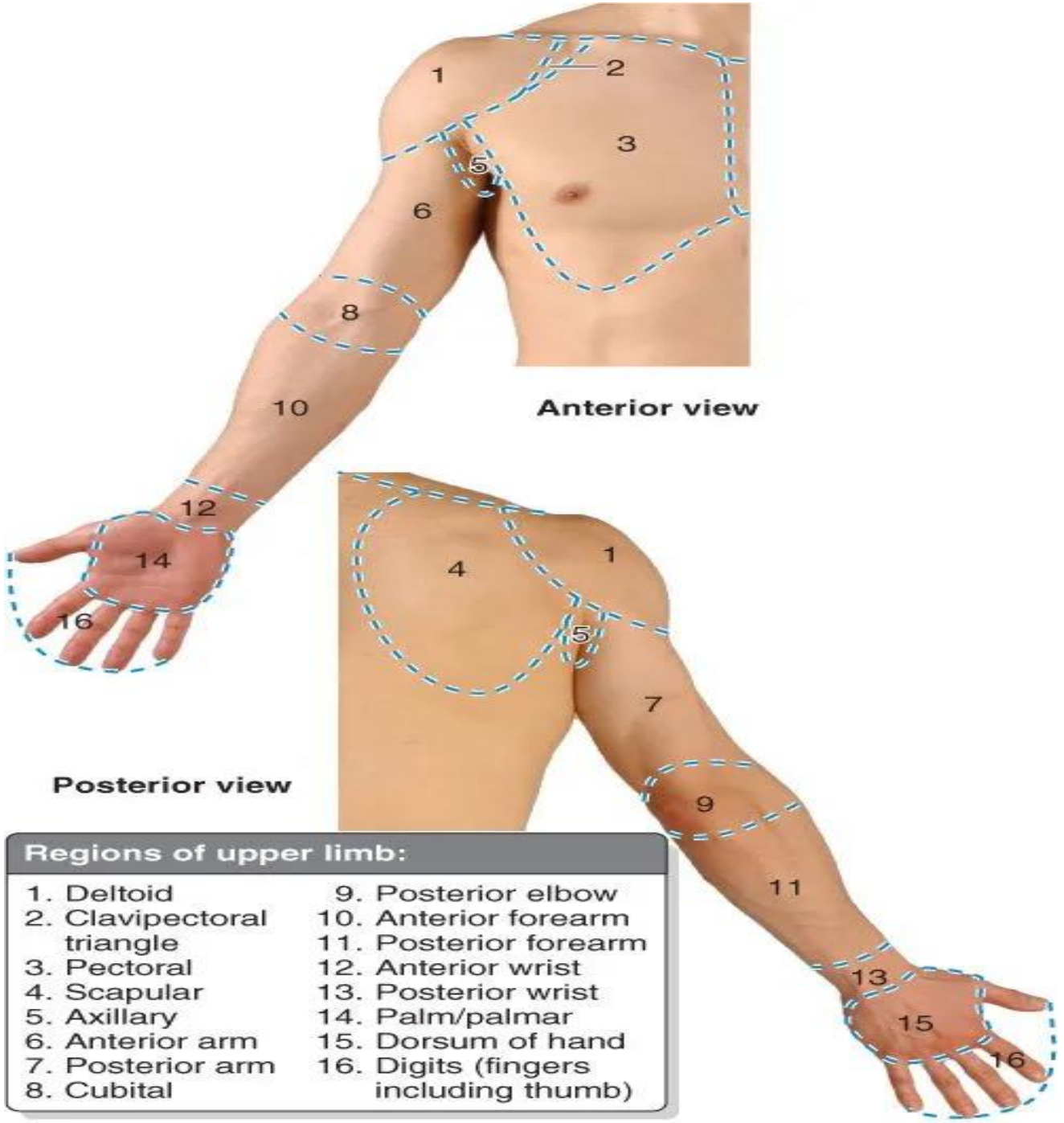
Ngozi Onuegbu PhD and Walaa Abu-Taleb

Learning Outcomes

By the end of this lecture, you will be able to:

- Accurately identify and palpate major anatomical landmarks of the upper limb
- Understand the functional significance of various bony prominences
- Describe the surface anatomy and palpation techniques for identifying structures of the upper limb
- Differentiate Between Upper Limb Joint Landmarks
- Apply the knowledge of anatomical landmarks in a clinical setting

Regions of the Upper Limb



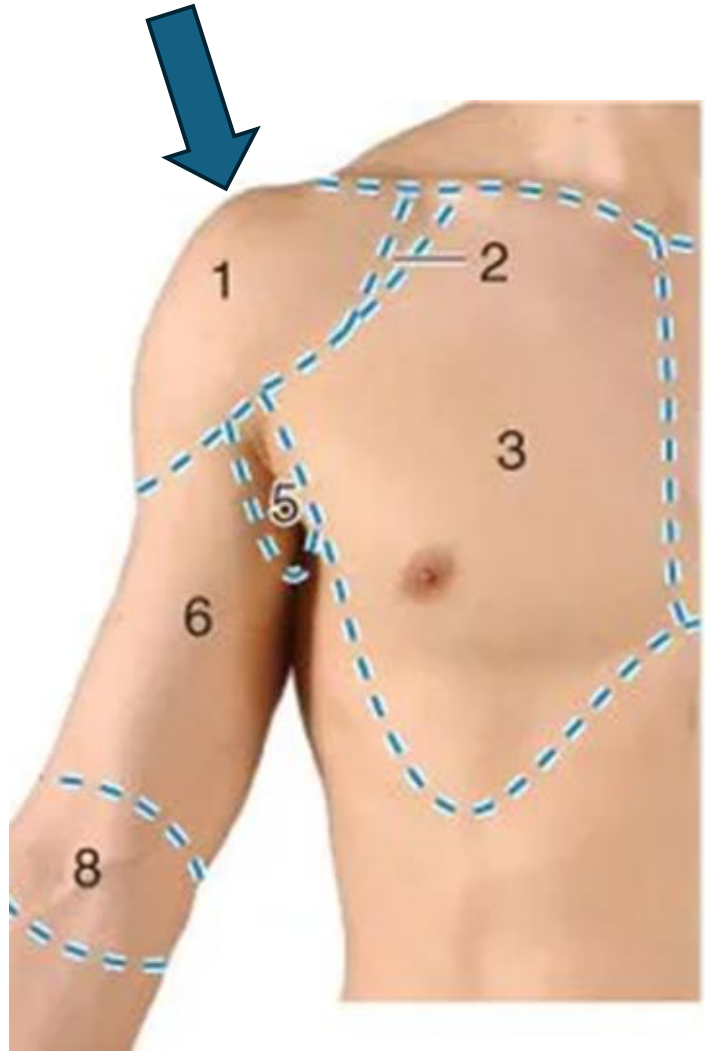
Deltoid Region

- The lateral aspect of the shoulder, covering the deltoid muscle. It forms the rounded contour of the shoulder.

Key Anatomical Features:

- 📍 **Boundaries:**
- **Superiorly:** Clavicle, acromion of the scapula
- **Inferiorly:** Deltoid tuberosity of the humerus
- **Anteriorly:** Overlaps with the clavipectoral and pectoral regions
- **Posteriorly:** Blends with the scapular region

- *In your groups, trace and identify these regions on your models*



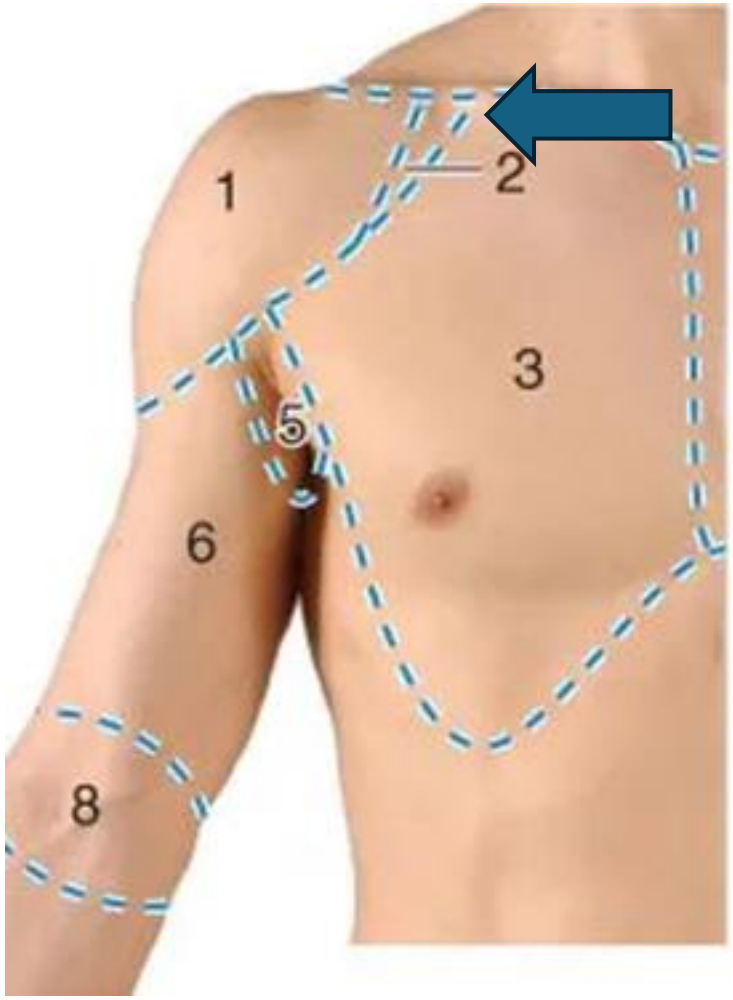
Clavipectoral Region of the Upper Limb: Surface Anatomy

- The clavipectoral region is a part of the anterior shoulder and upper chest area, located deep to the pectoralis major muscle and extending from the clavicle to the upper part of the axilla (armpit).

Boundaries:

- **Superiorly:** Clavicle
- **Inferiorly:** Upper border of pectoralis minor
- **Medially:** First rib
- **Laterally:** Coracoid process of the scapula

In your groups, trace and identify these regions on your models



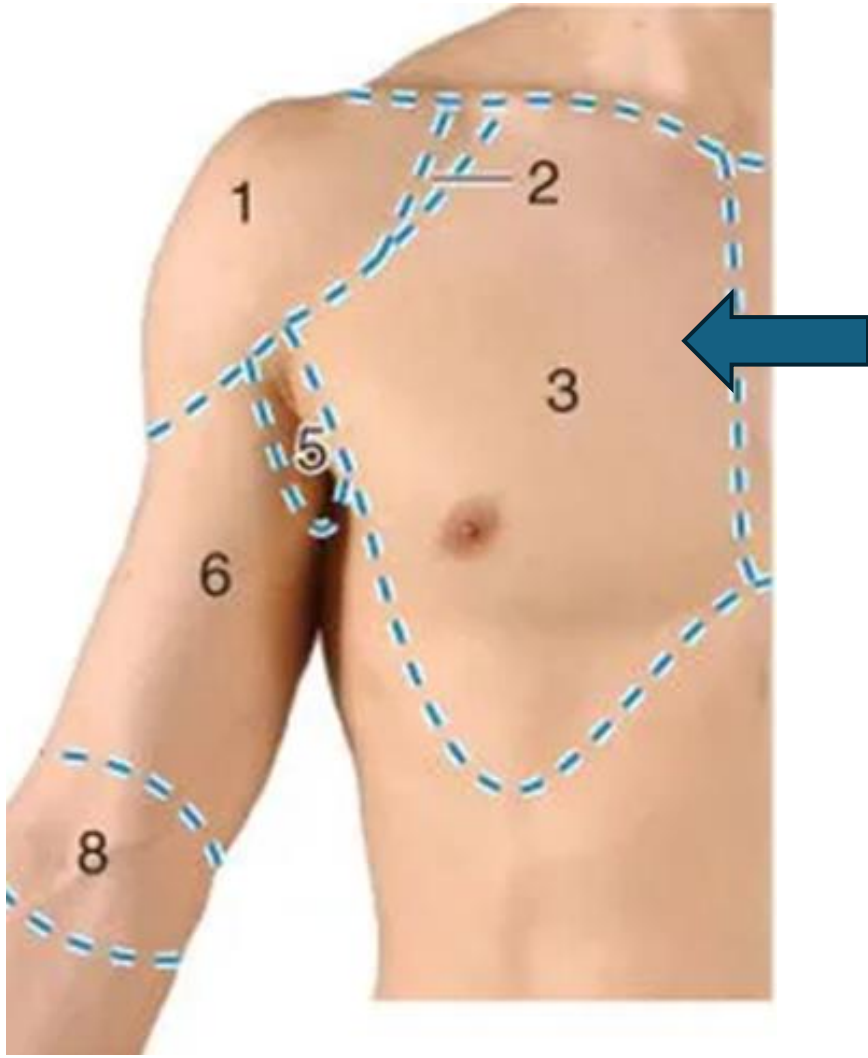
Pectoral Region

- The pectoral region refers to the anterior chest wall, particularly the area covering the upper thorax and shoulder.

Key Anatomical Features:

- 📍 **Boundaries:**
- **Superiorly:** Clavicle
- **Inferiorly:** Sixth or seventh rib
- **Medially:** Sternum
- **Laterally:** Transition into the axillary and deltoid regions

In your groups, trace and identify these regions on your models



Scapular Region

- The scapular region refers to the posterior aspect of the shoulder girdle, centred around the scapula (shoulder blade).

Key Anatomical Features:

- 📍 **Boundaries:**

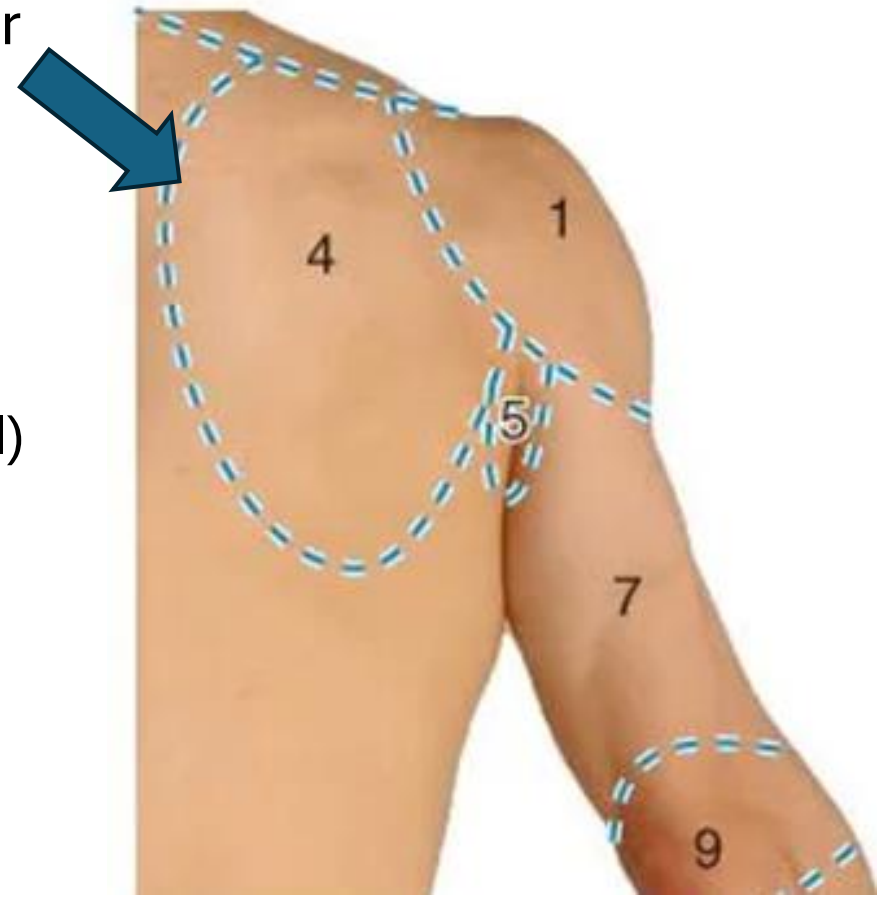
•**Superiorly:** Level of the second rib and spine of the scapula

•**Inferiorly:** Inferior angle of the scapula (around the 7th rib level)

•**Medially:** Vertebral column

•**Laterally:** Shoulder joint and deltoid region

In your groups, trace and identify these regions on your models



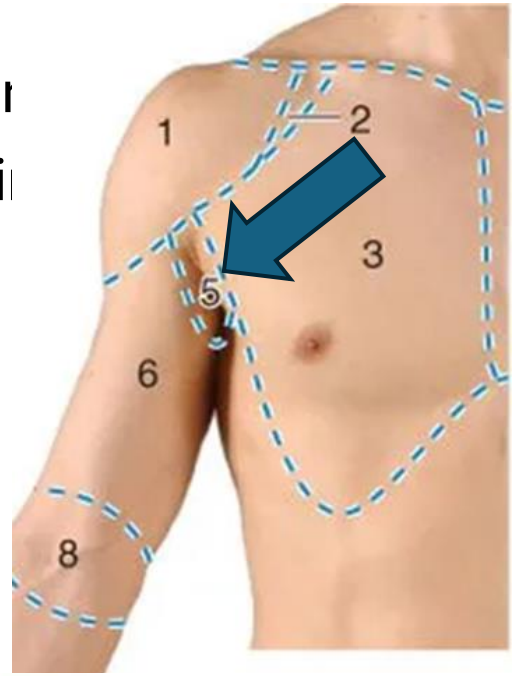
Axillary Region The axillary region (commonly referred to as the ar

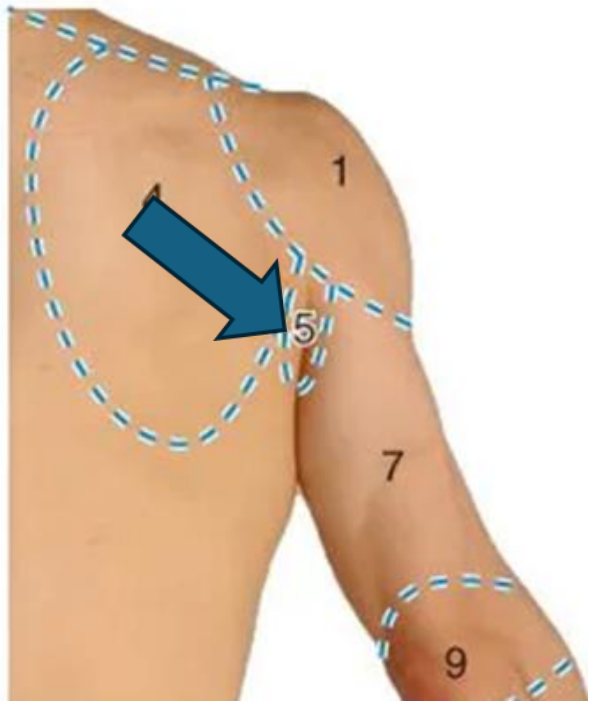
a pyramidal space located between the upper thoracic wall and the upper li

Key Anatomical Features:

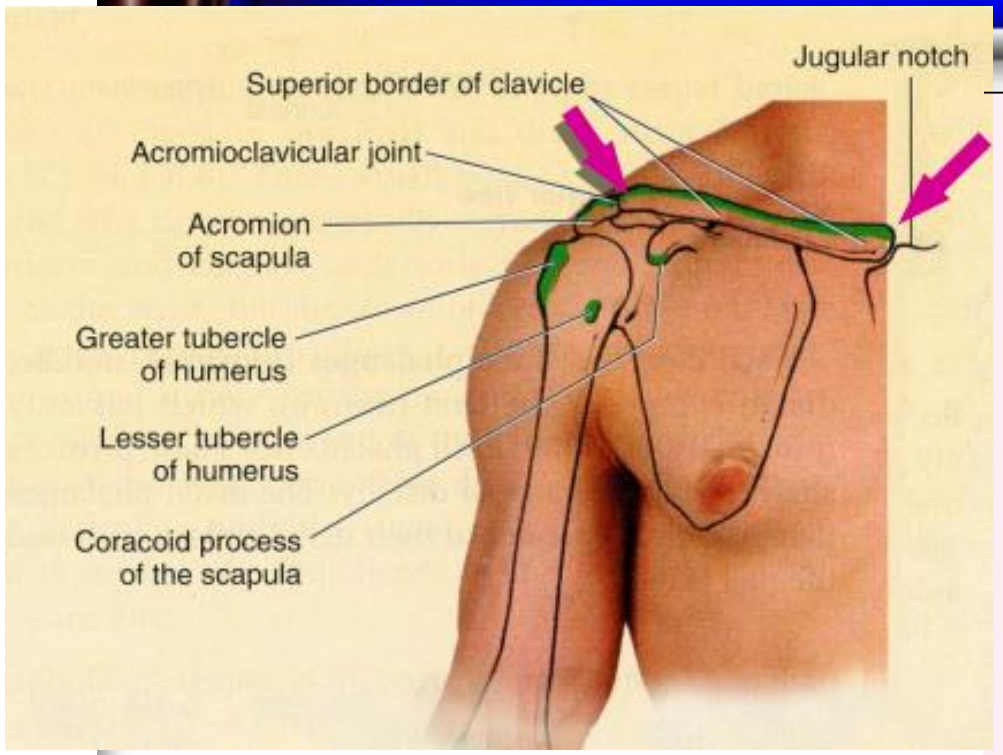
- 📍 **Boundaries:** *In your groups, trace and identify these regions on your models*

Wall	Structures
Apex (inlet)	Bounded by the clavicle (anterior), first rib (medial), and superior border of scapula (posterior) – opens into the root of the neck
Base (floor)	Skin, subcutaneous tissue, and axillary fascia – forms the floor of the axilla
Anterior wall	Pectoralis major, pectoralis minor, and clavipectoral fascia
Posterior wall	Subscapularis, teres major, and latissimus dorsi
Medial wall	Upper thoracic wall (ribs 1–4) and serratus anterior
Lateral wall	Intertubercular groove of the humerus



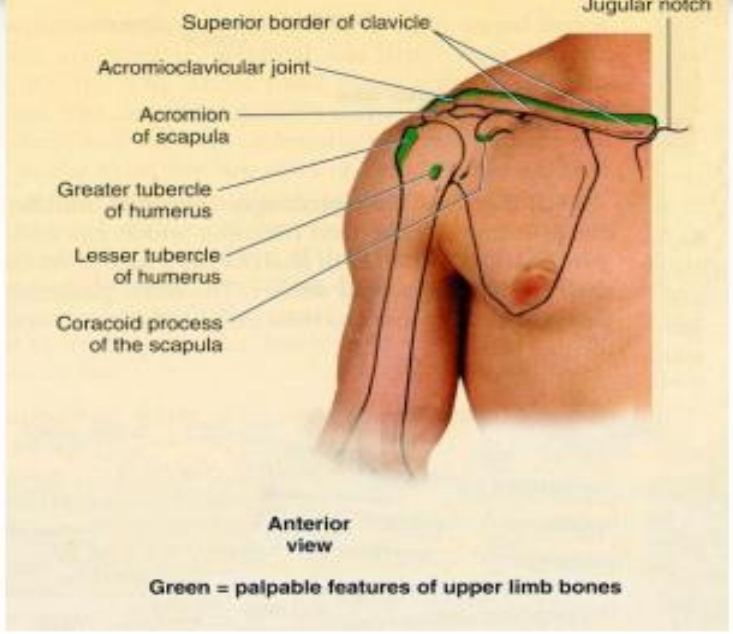


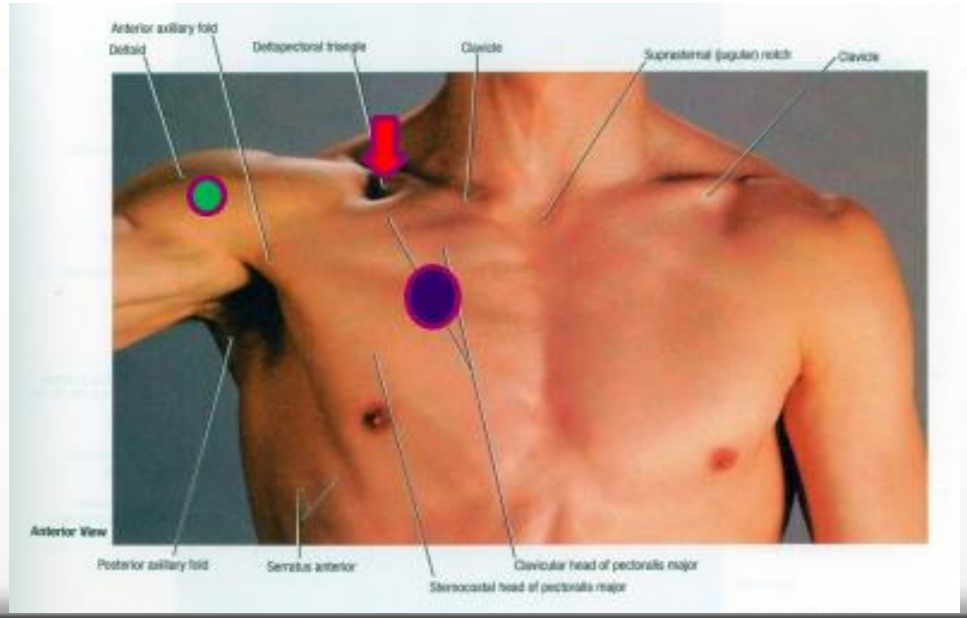
Important Landmarks of the Pectoral Region and Arm



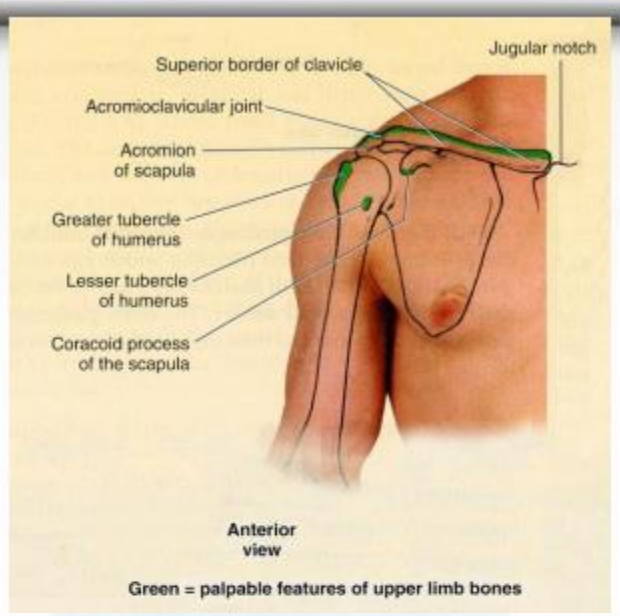


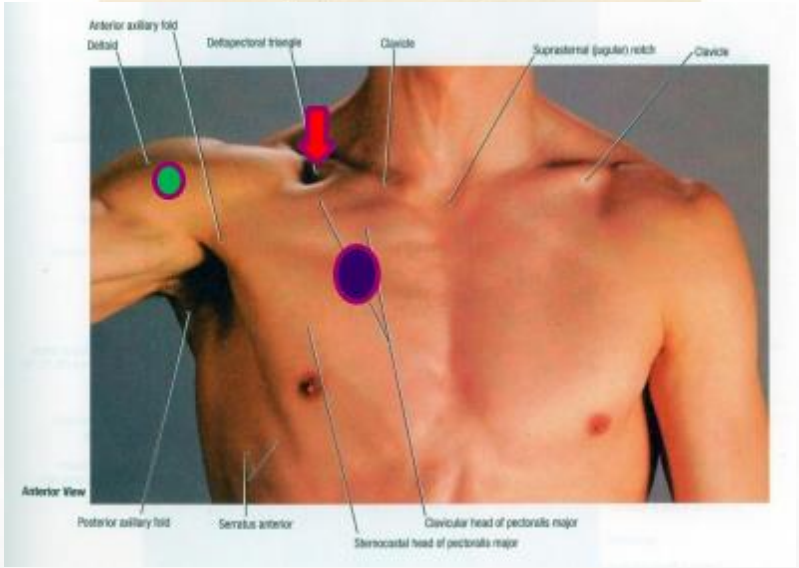
- The **clavicle** is subcutaneous and can be palpated throughout its length.
- Its **sternal end** projects little above the manubrium.
- Between the 2 sternal ends of the 2 clavicles lies the **jugular notch** (suprasternal notch).
- The **acromial** end of the clavicle can be palpated medial to the lateral border of the **acromion,** of the scapula. particularly when the shoulder is alternately raised and depressed.
- The large vessels and nerves to the upper limb pass posterior to the convexity of the **clavicle.**





- The **coracoid process** of scapula can be felt deeply below the lateral one third of the clavicle in the **Deltopectoral GROOVE or clavipectoral triangle.**
- The **clavipectoral or the (Deltopectoral) triangle** is the slightly depressed area just inferior to the lateral third of clavicle.
- The clavipectoral triangle is bounded by:
  - Clavicle **superiorly,**
  - Deltoid **laterally,** and
  - Pectoralis major **medially.**

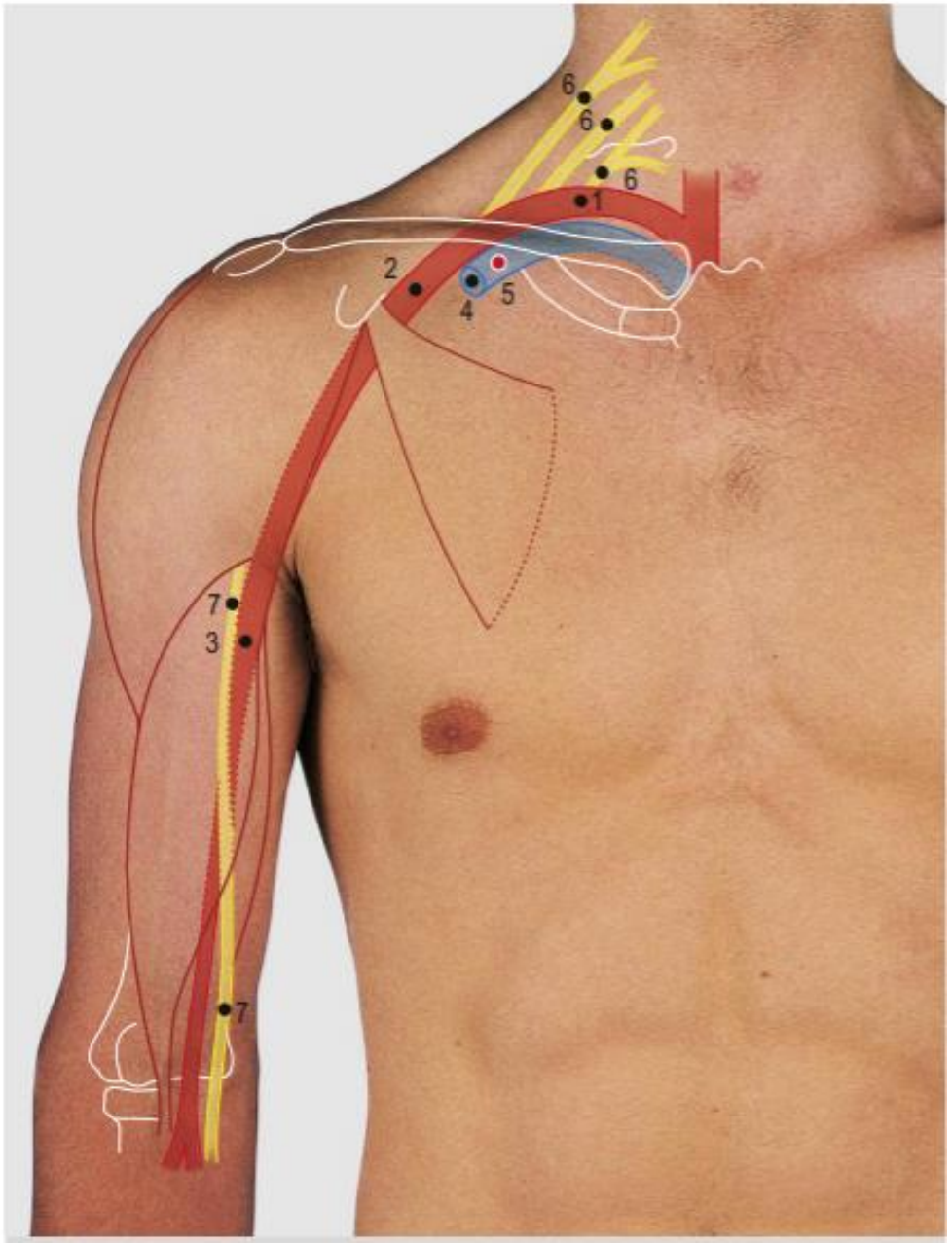




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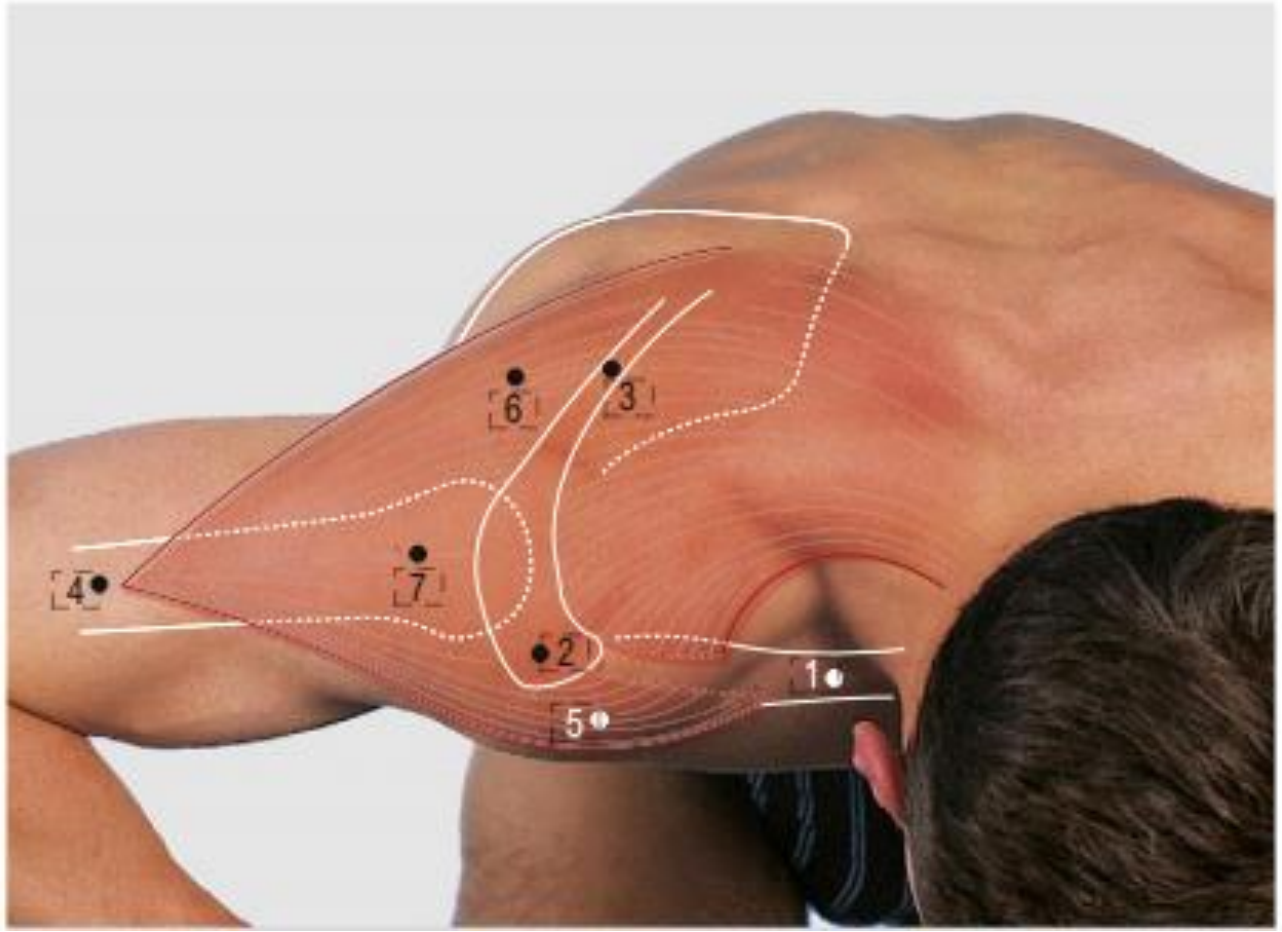
Anterior aspect of the shoulder and upper arm: vessels and nerves

1 Subclavian artery
2 Axillary artery
3 Brachial artery
4 Axillary vein
5 Site of access to subclavian vein
6 Brachial plexus: trunks separated for clarity
7 Median nerve



Attachments and action of the deltoid muscle

1 Clavicle
2 Acromion
3 Spine of the scapula
4 Humerus
5 Anterior fibres
6 Posterior fibres
7 Lateral fibres

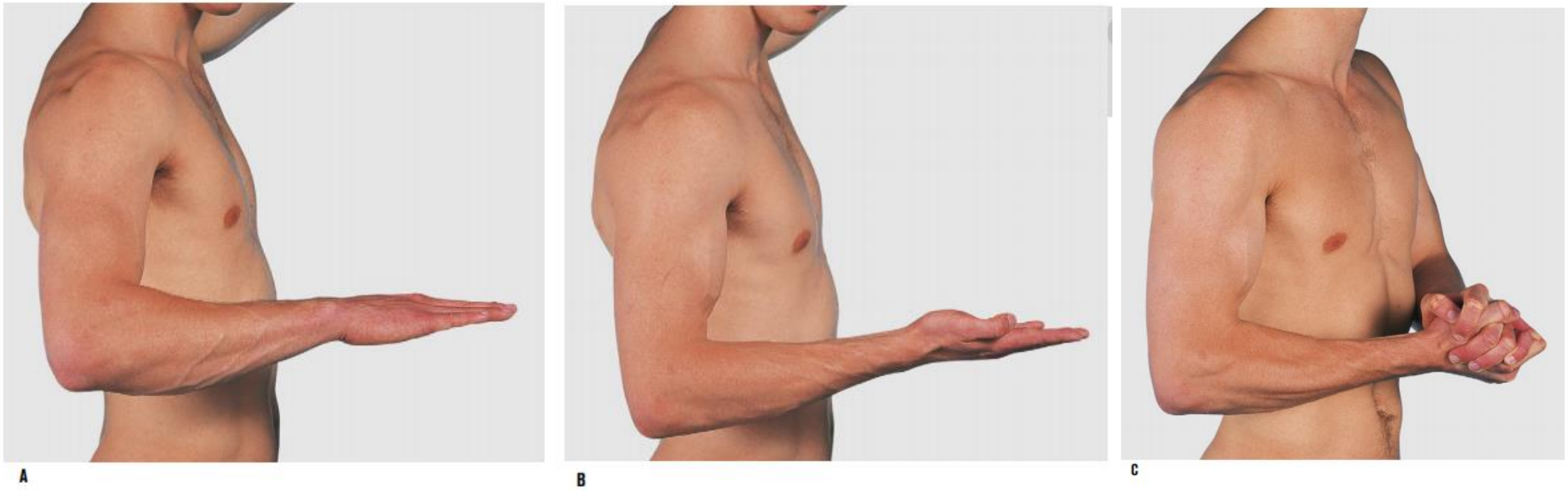


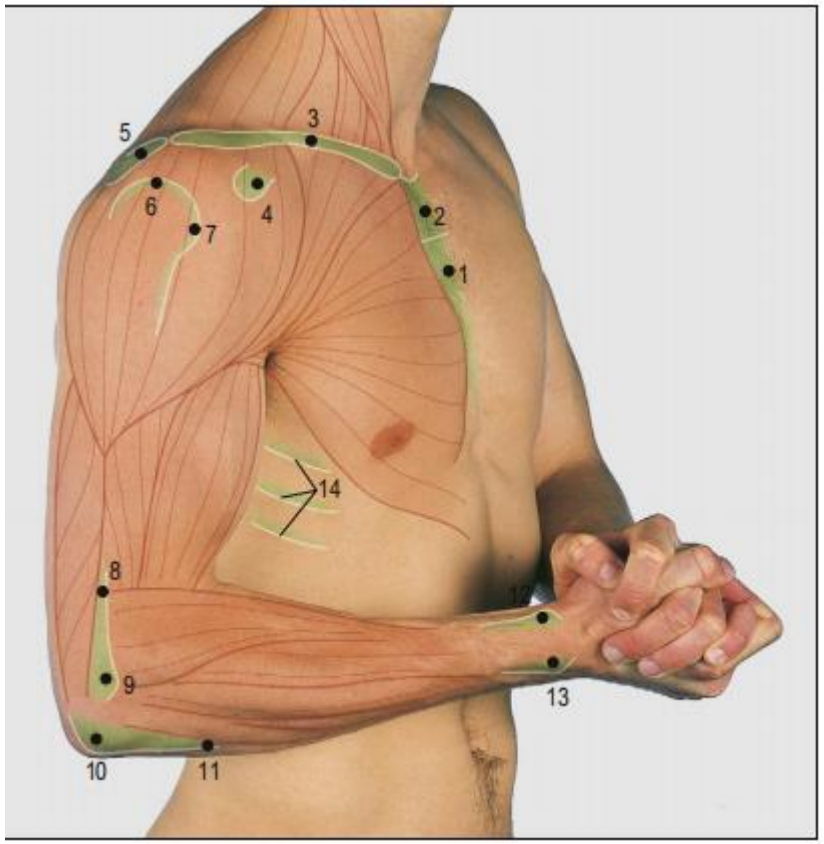
Actions of the biceps muscle

A, pronation;

B, supination;

C, forced flexion and supination



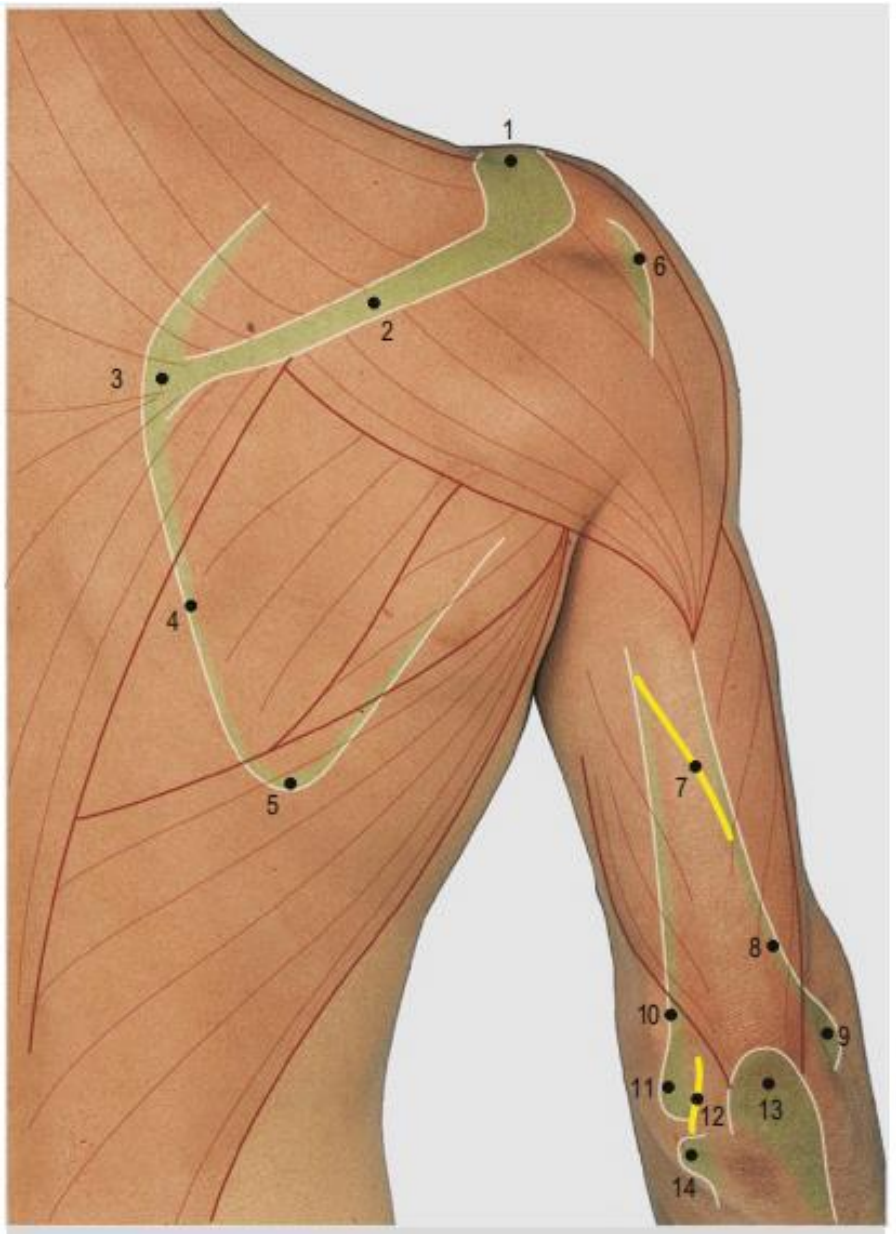


Lateral aspect of the upper limb: palpable structures

1 Body of the sternum
2 Manubrium sterni
3 Clavicle
4 Coracoid process
5 Acromion
6 Greater tuberosity
7 Lesser tuberosity
8 Lateral supracondylar ridge
9 Lateral epicondyle
10 Olecranon
11 Subcutaneous border of the ulna
12 Radial tuberosity
13 Head of the ulna
14 Lateral aspect of ribs

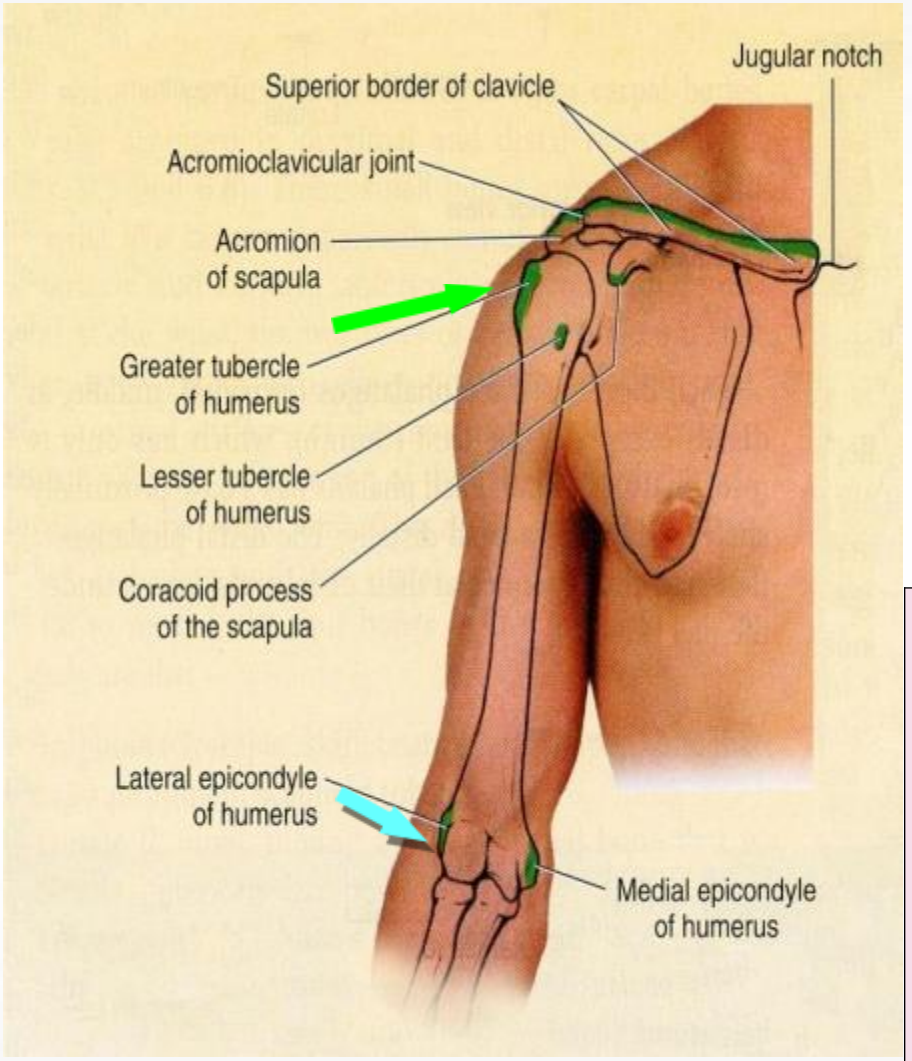


7.13 Posterior aspect of the shoulder and upper arm



Posterior aspect of shoulder and upper arm: palpable structures

1 Acromion
2 Spine of scapula
3 Medial angle
4 Medial border
5 Inferior angle of scapula
6 Greater tuberosity
7 Radial nerve deep to fibres of triceps
8 Lateral supracondylar ridge
9 Lateral epicondyle
10 Medial supracondylar ridge
11 Medial epicondyle
12 Ulnar nerve
13 Olecranon process
14 Head of ulna

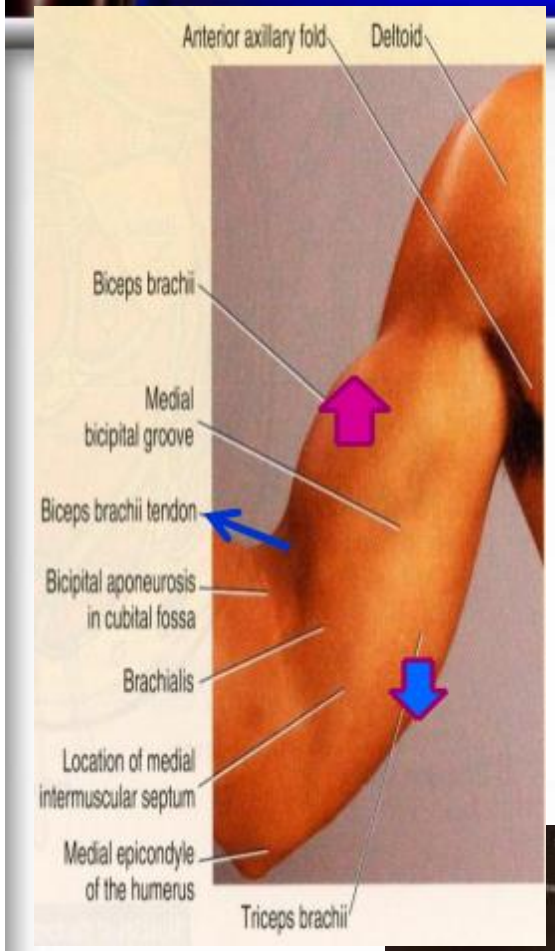


- The **greater tubercle of humerus** can be felt by deep palpation through the deltoid muscle, inferior to the acromion when the arm is by the side.
- In this position, the greater tubercle is the most lateral bony point of the shoulder.

- The **shaft of the humerus** may be felt in different areas through the muscles surrounding it.
- The **medial** and **lateral epicondyles of the humerus** are palpated on the medial & lateral sides of the elbow region.

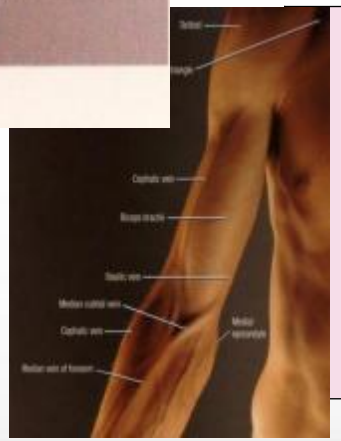
- **Axillary Folds**
- **The anterior axillary fold** is formed by the lower margin of the pectoralis major, and can be palpated by the finger.
- This can be made by asking the patient to press his or her hand against the ipsilateral hip.
- The posterior axillary fold is formed by the tendons of latissimus dorsi and teres major muscles
- **Axilla**
- The axilla should be examined with the forearm supported and the pectoral muscles relaxed.
- When the arm by the side, the inferior part of the head of the humerus can be easily palpated through the floor of the axilla.
- The pulsations of the **axillary artery** can be felt high up in the axilla, and around the artery the cords of the brachial plexus.
- The medial wall of the axilla is formed by the upper ribs covered by the serratus anterior.
- The lateral wall is formed by the coracobrachialis and biceps brachii and the bicipital groove.





- The borders of the **deltoid** are visible when the arm is abducted against resistance.
- The **distal attachment of the deltoid** can be palpated on the lateral surface of the humerus.

- Biceps brachii & triceps brachii form a bulge on the anterior and posterior surfaces of the arm.
- The **biceps tendon** can be palpated in the cubital fossa, immediately lateral to the midline.
- The triceps tendon can be palpated where it is attached to the olecranon process.



There are 2 grooves: Medial and lateral grooves separate the bulges formed by the biceps and triceps.

The **cephalic vein** ascends superiorly in the lateral groove and

The **basilicvein** ascends in the medial groove.

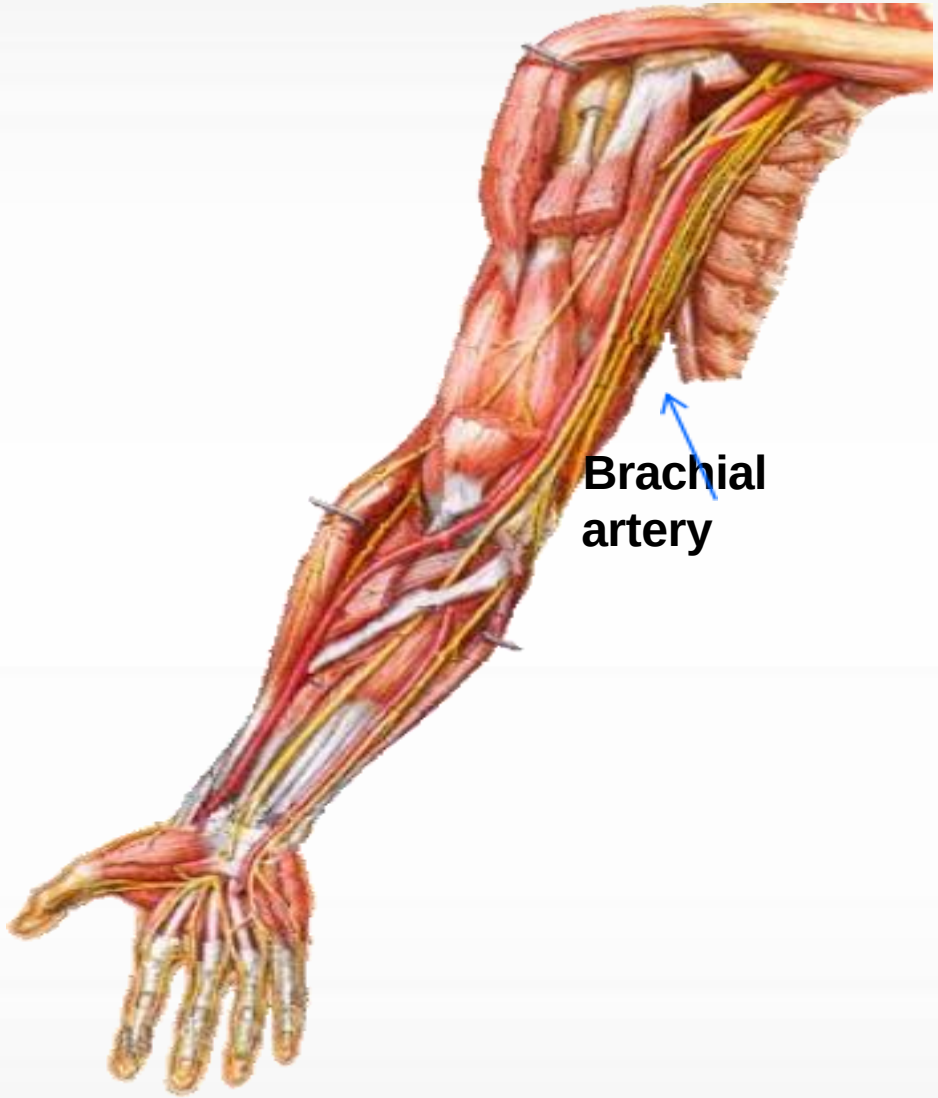
The **brachial artery** can be felt pulsating deep to the medial border of the biceps.

To stop bleeding by pressure on the artery in the upper half of the arm it is pushed **laterally** against the humerus.

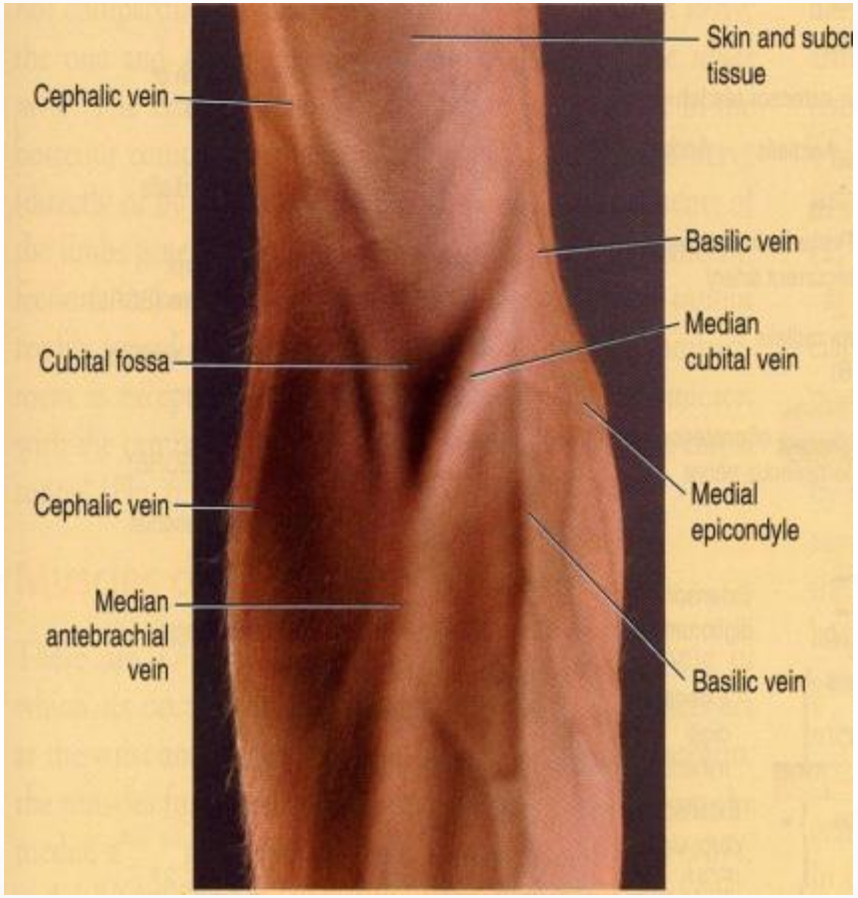
In the lower half it is pushed **posteriorly**.

In the cubital fossa, it lies beneath the bicipital aponeurosis.

At the level of the neck of the radius, it divides into radial and ulnar arteries.



CUBITAL FOSSA



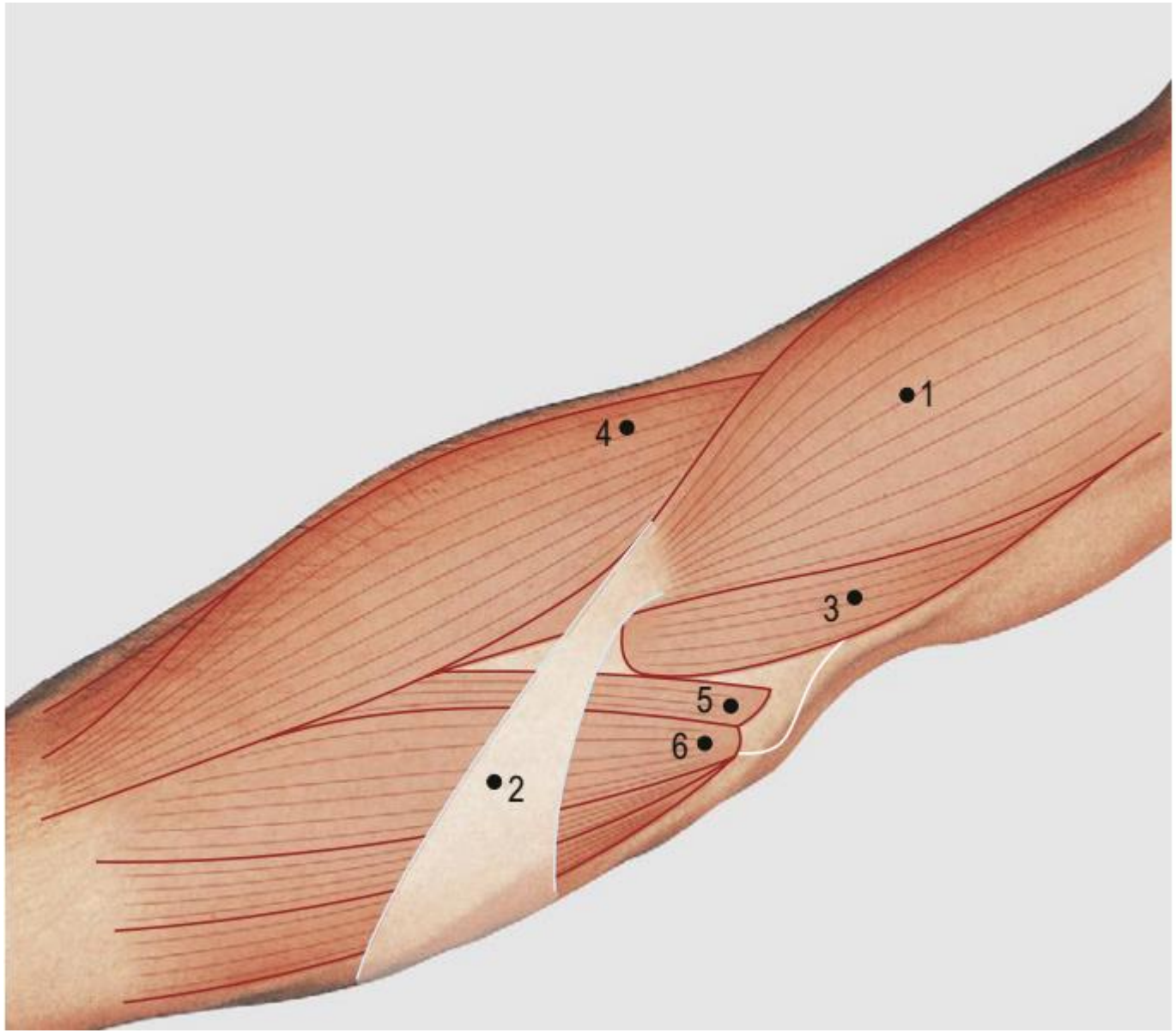
In the cubital fossa, try to locate:

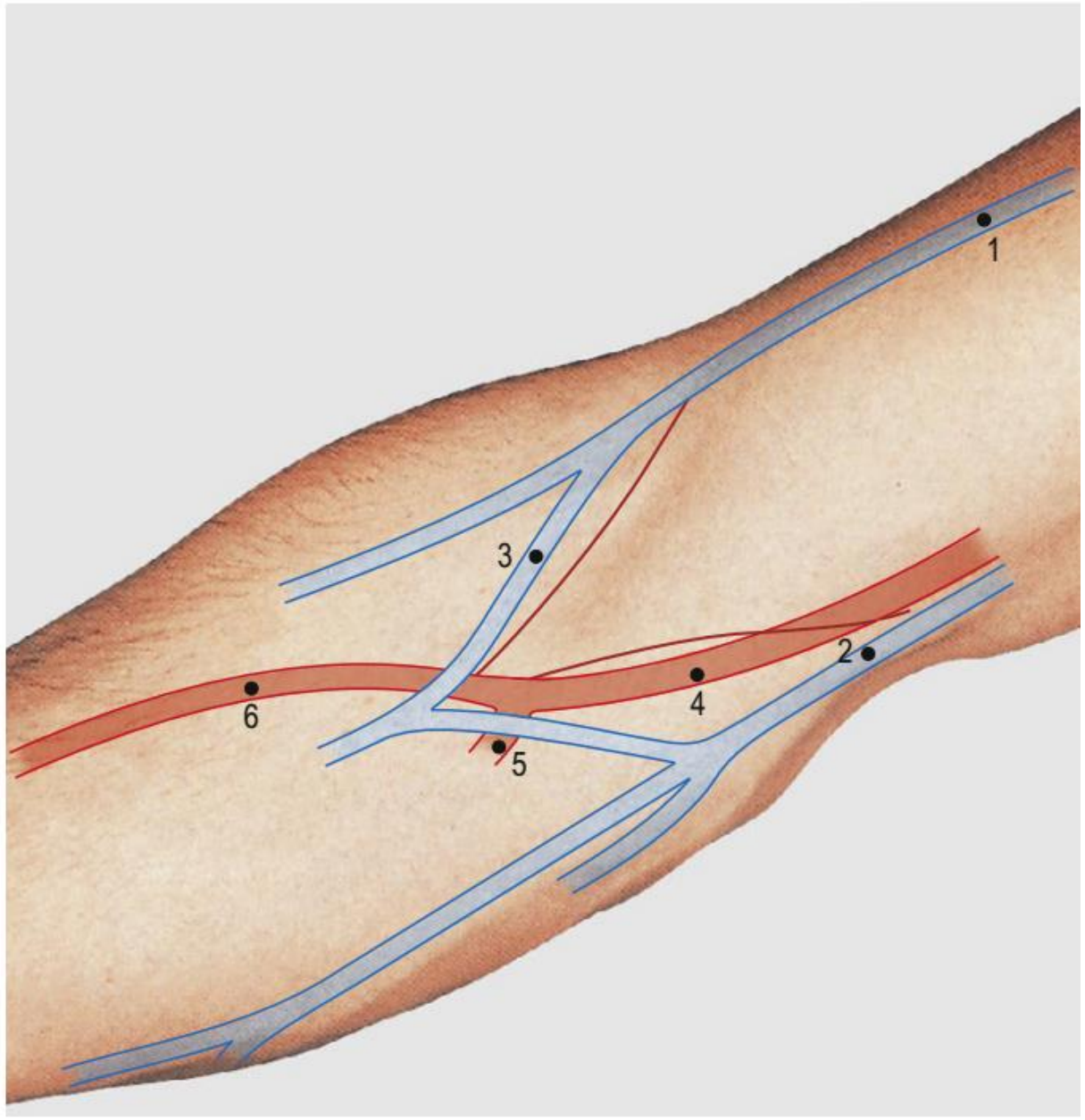
- **Cephalic vein**
- **Basilic vein** and
- M**edian cubital vein** are clearly visible.
- The median cubital vein connects the cephalic and the basilic veins .
- **It crosses over the bicipital aponeurosis.**
- It is the vein of choice for IV line, **WHY?**

Cubital Fossa

1 Biceps
2 Bicipital aponeurosis
3 Brachialis
4 Brachioradialis
5 Pronator teres
6 Common flexor origin

Identify the superficial muscles of the forearm in your groups





Cubital Fossa

1 Cephalic vein

2 Basilic vein

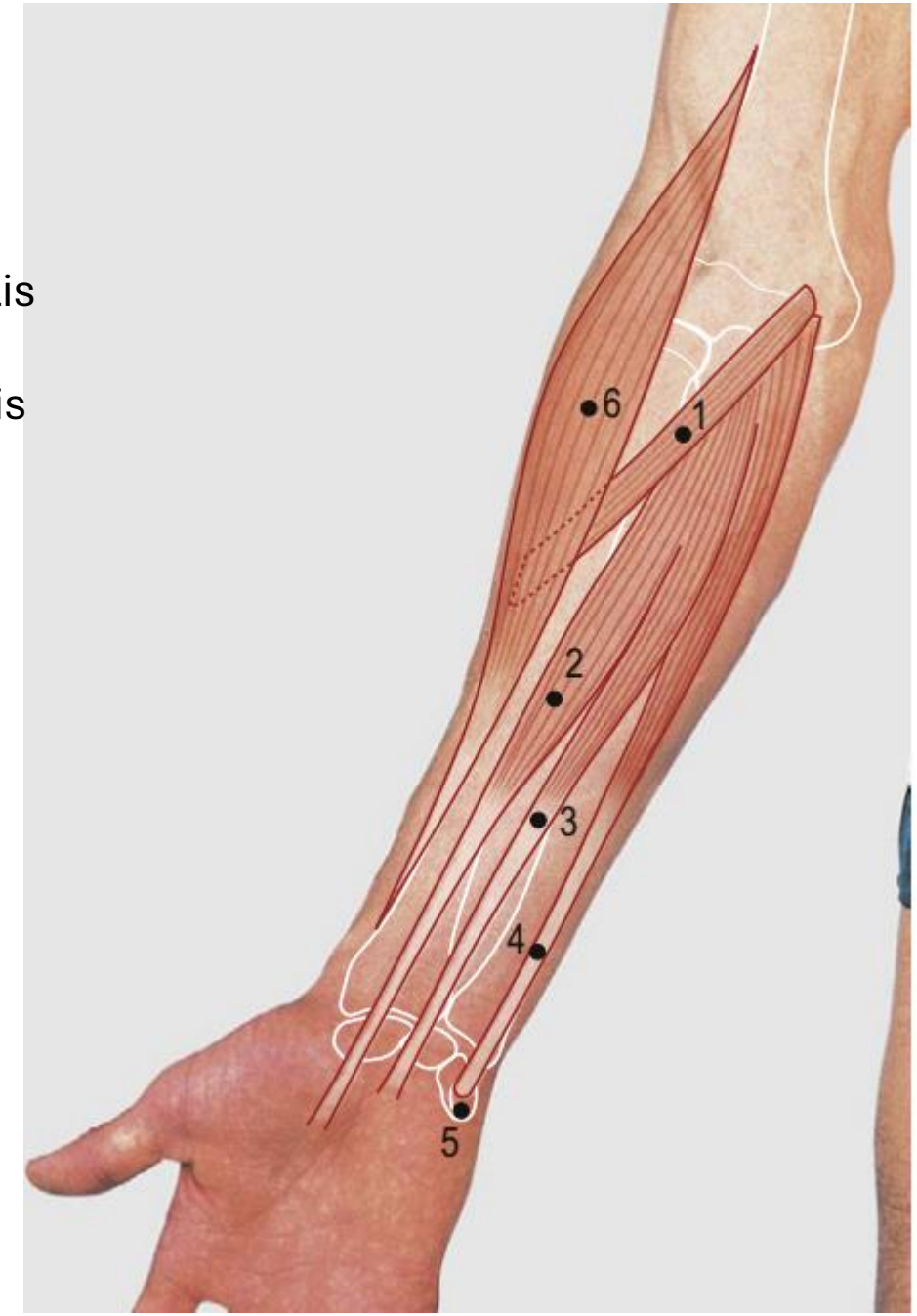
3 Median cubital vein

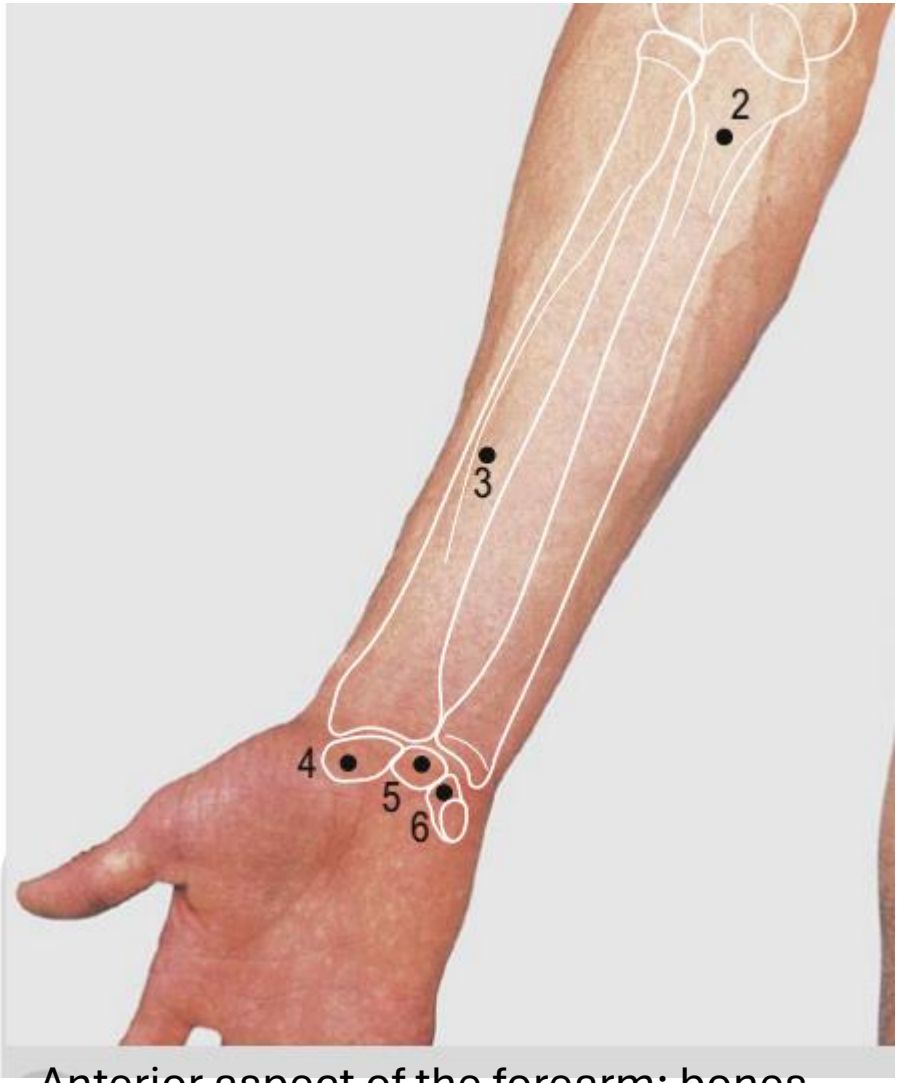
4 Brachial artery

5 Ulnar artery

6 Radial artery

1 Pronator teres
2 Flexor carpi radialis
3 Palmaris longus
4 Flexor carpi ulnaris
5 Pisiform bone
6 Brachioradialis





Anterior aspect of the forearm: bones
2 Ulna
3 Radius
4 Scaphoid
5 Lunate
6 Triquetral



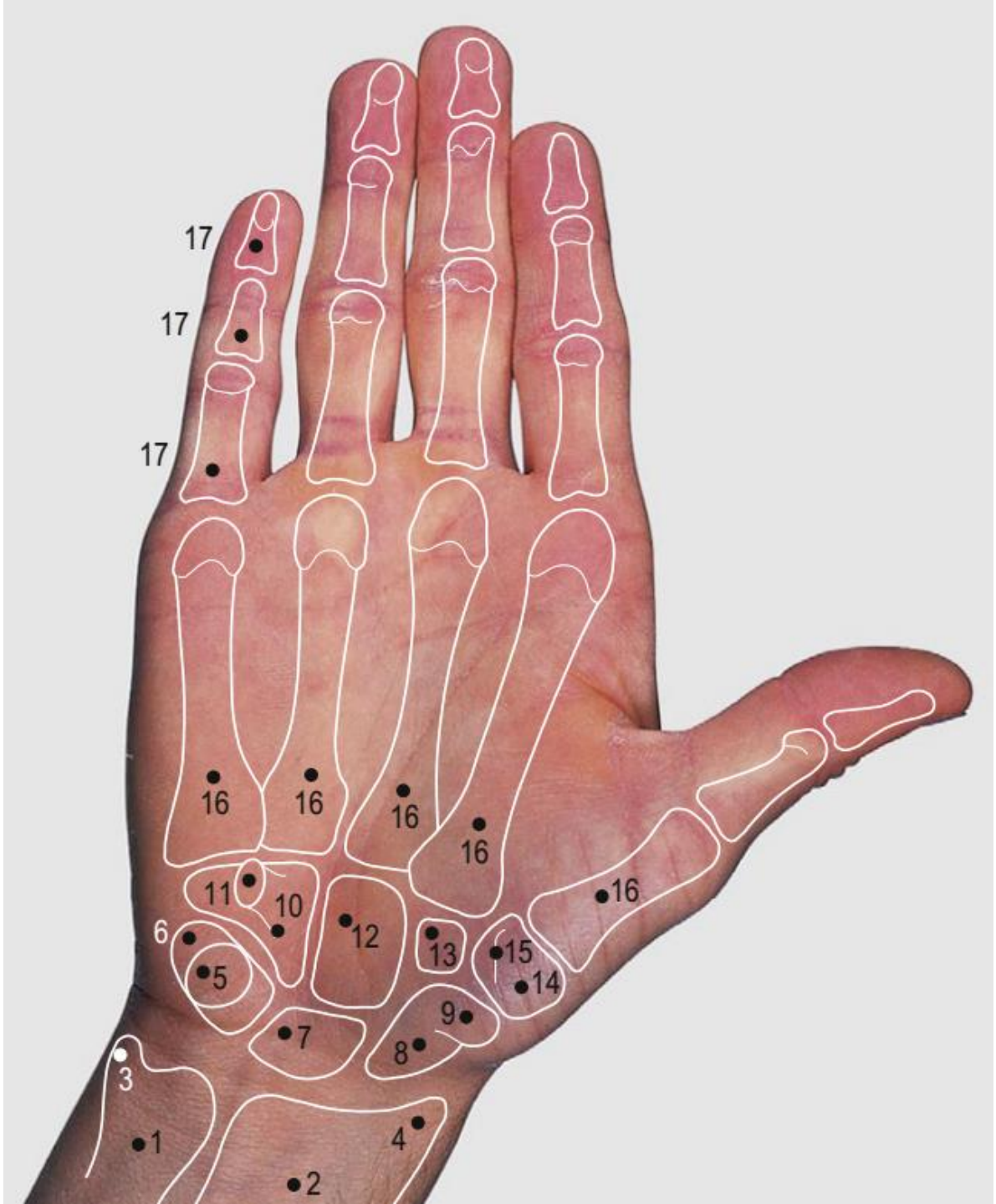
1 Brachioradialis
2 Extensor carpi radialis longus
3 Anconeus
4 Extensor carpi radialis brevis
5 Extensor digitorum
6 Extensor carpi ulnaris
7 Extensor digiti minimi
8 Abductor pollicis longus
9 Extensor pollicis brevis
10 Extensor pollicis longus
11 Ulnar nerve
12 Flexor carpi ulnaris



1 Lateral supracondylar ridge
2 Lateral epicondyle
3 Head of radius
4 Olecranon process
5 Subcutaneous border of ulna
6 Radial styloid process
7 Ulnar styloid process

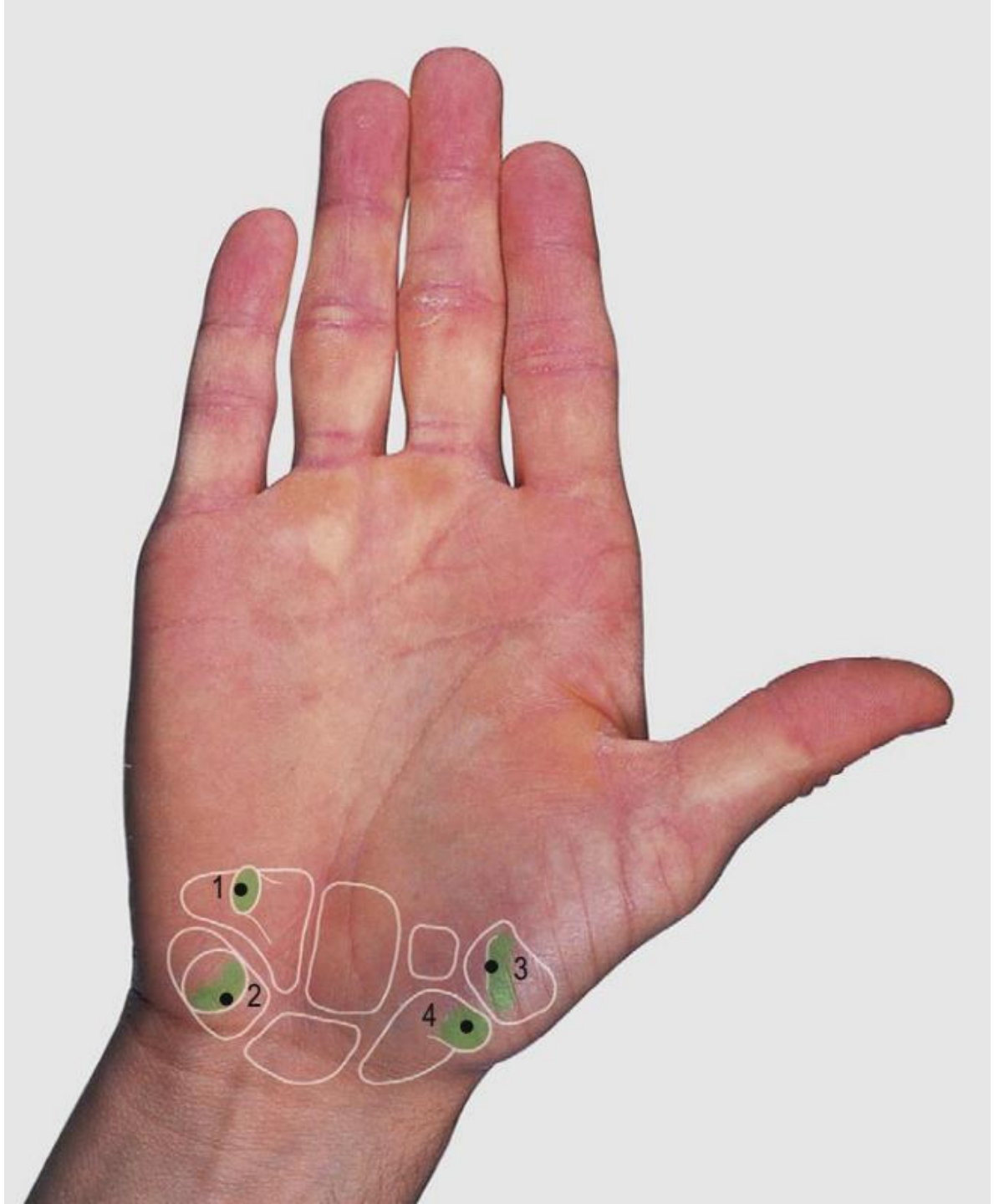
Forearm in full supination and pronation





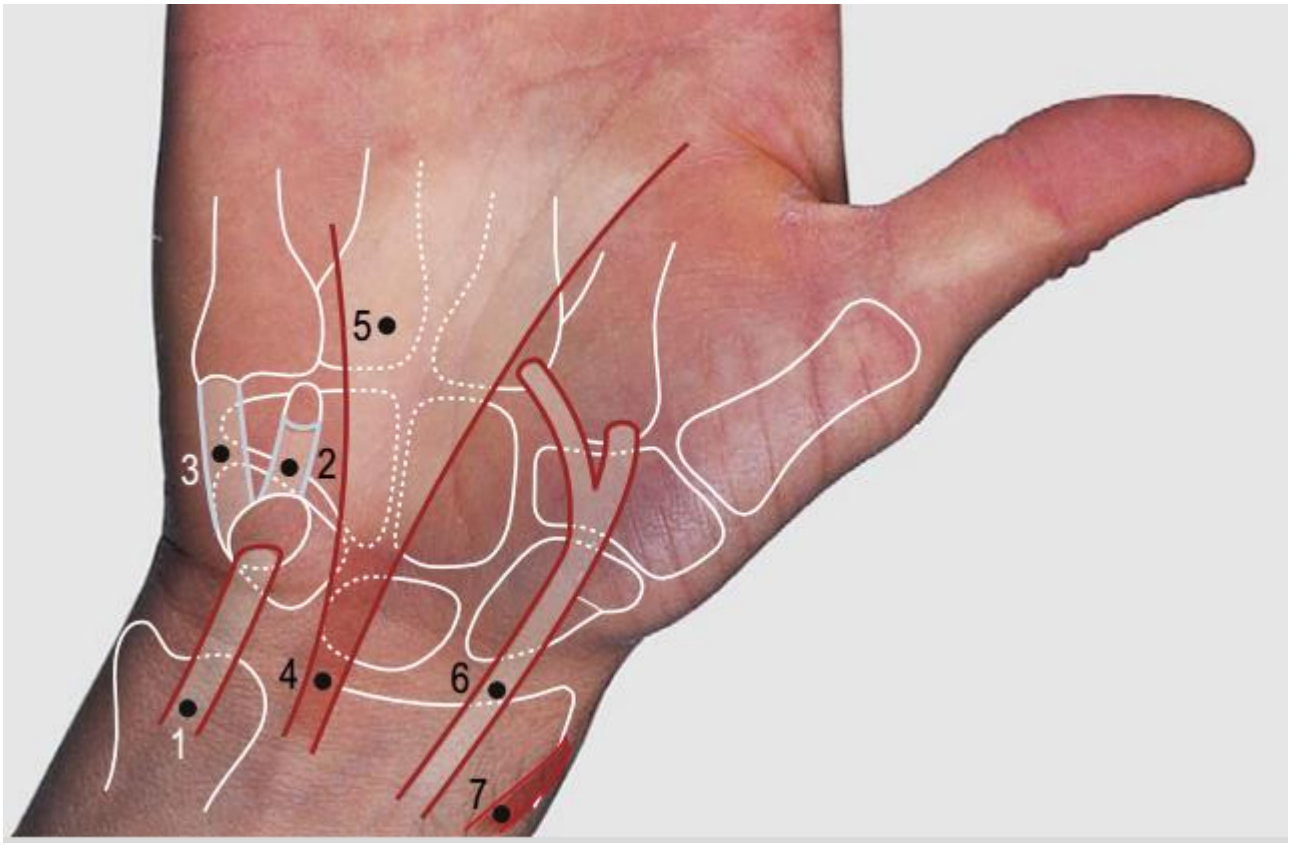
Hand bones

1 Ulna
2 Radius
3 Ulnar styloid process
4 Radial styloid process
5 Pisiform
6 Triquetral
7 Lunate
8 Scaphoid
9 Scaphoid tubercle
10 Hamate
11 Hook of hamate
12 Capitate
13 Trapezoid
14 Trapezium
15 Ridge of trapezium
16 Metacarpals 1–5
17 Phalanges



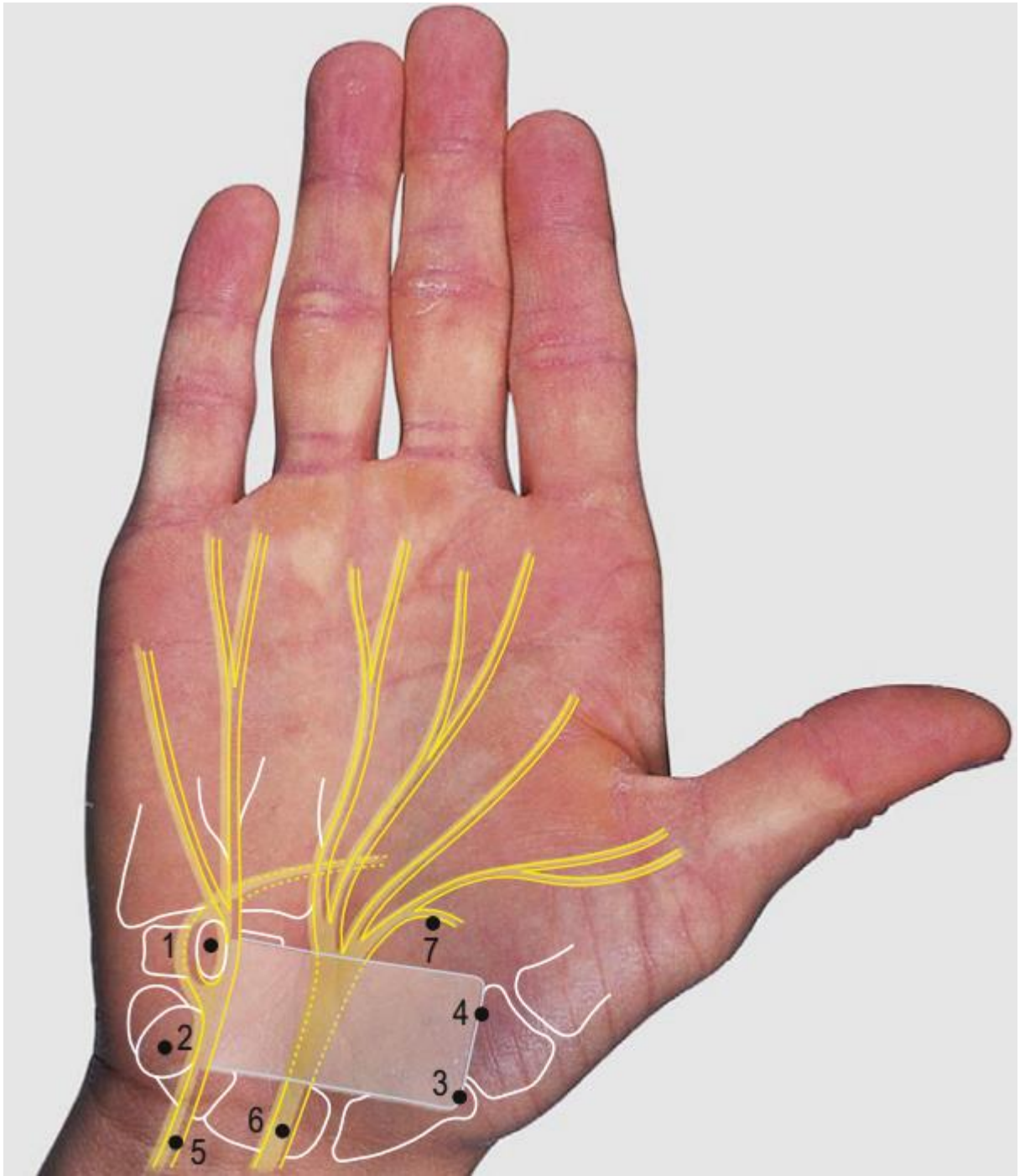
Flexor retinaculum: palpable attachments

1 Hook of hamate
2 Pisiform
3 Ridge of trapezium
4 Scaphoid tubercle



Anterior aspect of the wrist and hand: superficial tendons

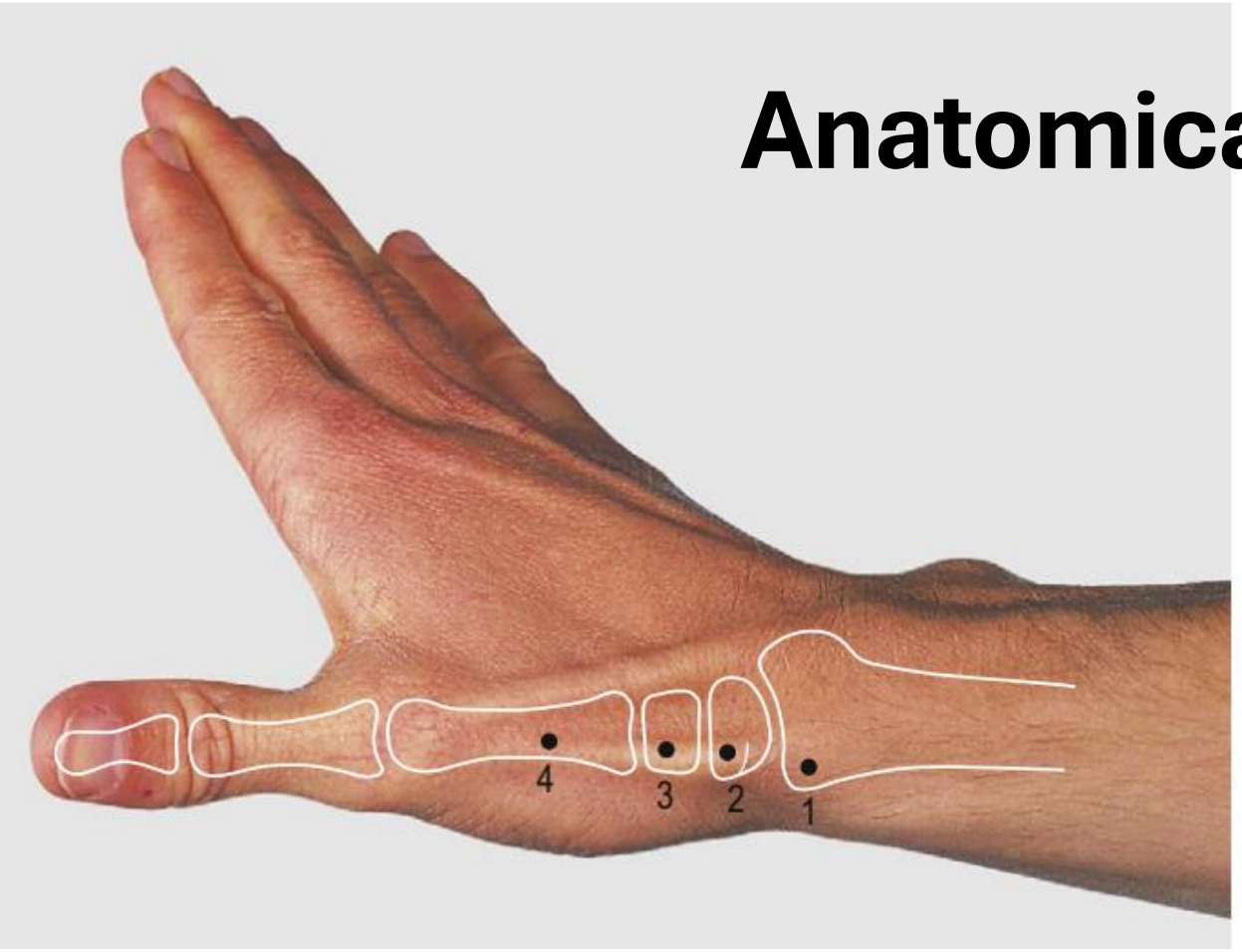
1 Flexor carpi ulnaris
2 Pisohamate ligament
3 Pisometacarpal ligament
4 Palmaris longus
5 Palmar aponeurosis
6 Flexor carpi radialis
7 Radial artery



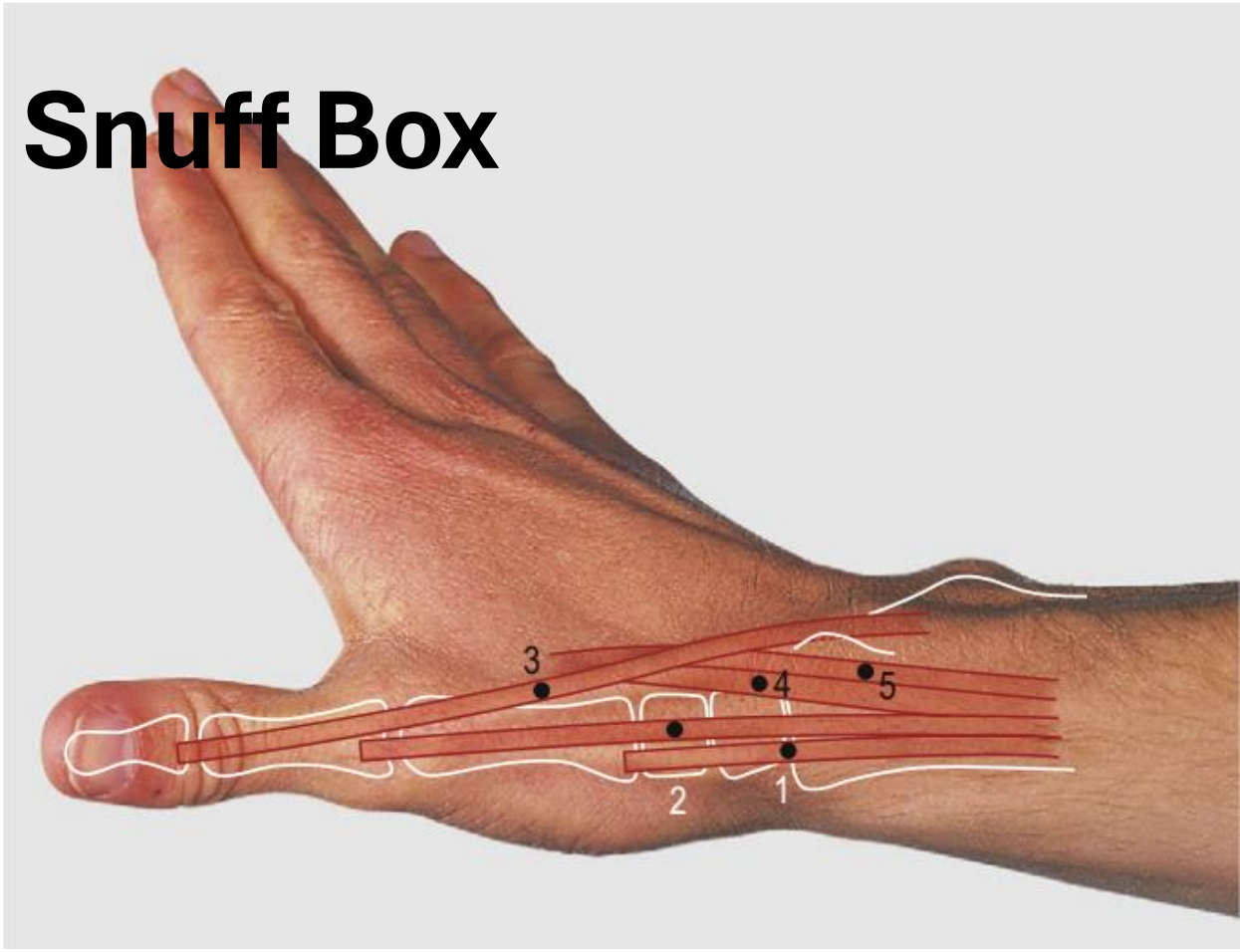
Flexor retinaculum

1 Hook of hamate
2 Pisiform bone
3 Scaphoid tubercle
4 Ridge of trapezium
5 Ulnar nerve
6 Median nerve
7 Recurrent branch of median nerve

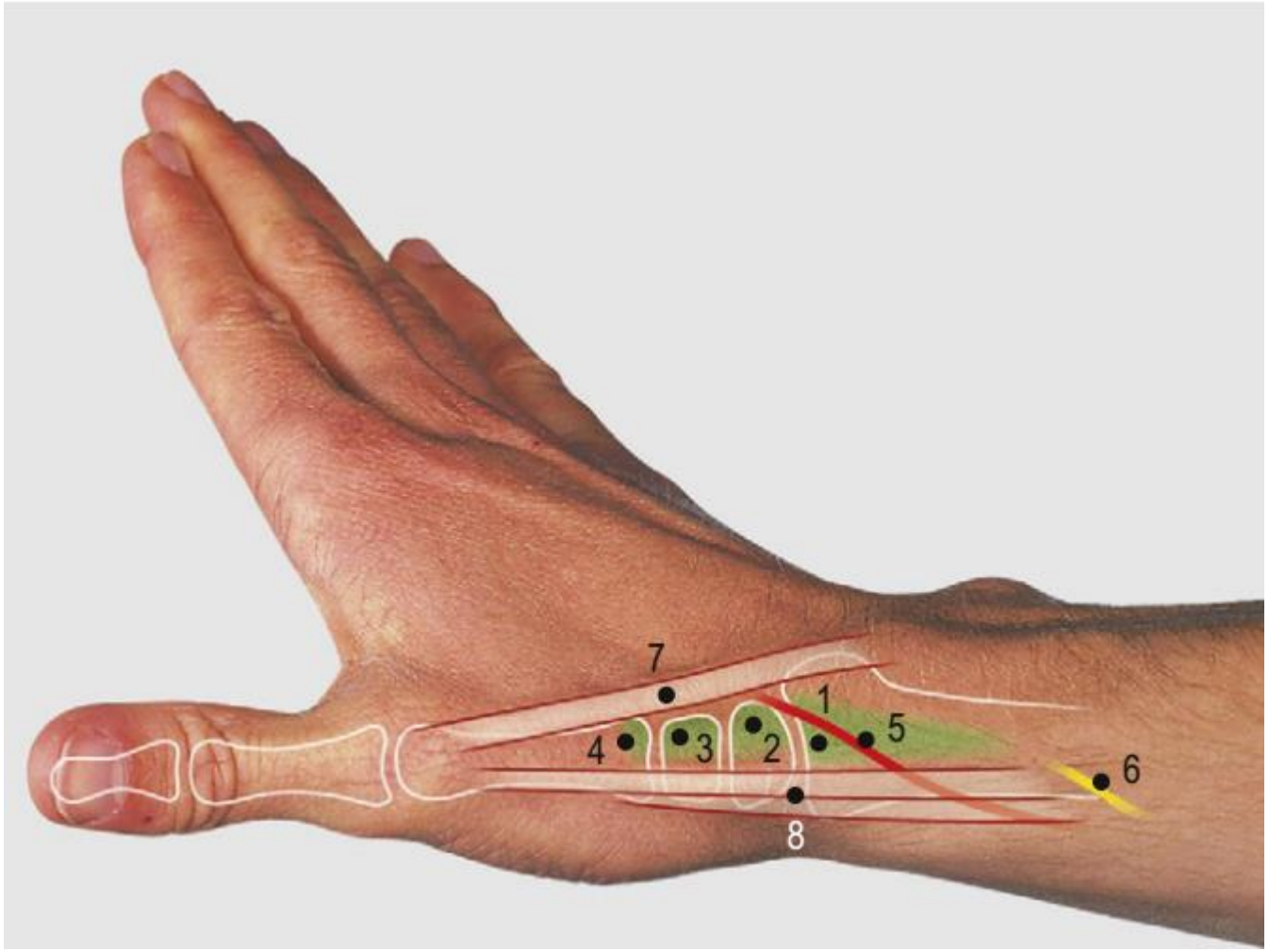
Anatomical Snuff Box



1 Radial styloid
2 Scaphoid
3 Trapezium
4 First metacarpal



1 Abductor pollicis longus
2 Extensor pollicis brevis
3 Extensor pollicis longus
4 Extensor carpi radialis longus
5 Extensor carpi radialis brevis



Anatomical Snuff Box

1 Radial styloid process
2 Scaphoid
3 Trapezium
4 Base of first metacarpal
5 Radial artery
6 Radial nerve
7 Extensor pollicis longus
8 Extensor pollicis brevis and abductor pollicis longus

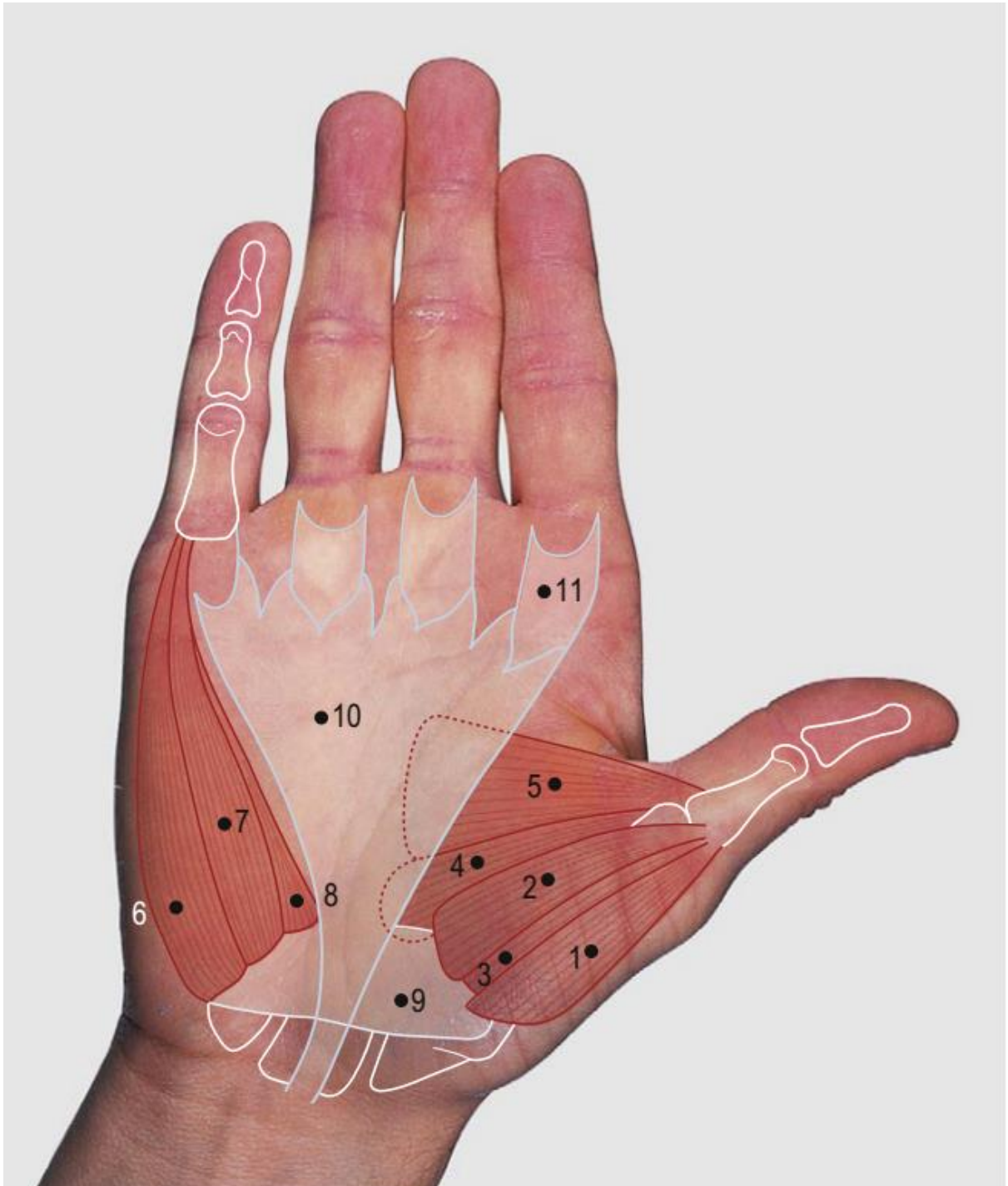
Bones of the back of the hand

1 Ulna
2 Ulna styloid
3 Radius
4 Radial styloid
5 Dorsal tubercle of radius
6 Triquetral
7 Lunate
8 Scaphoid
9 Hamate
10 Capitate
11 Trapezoid
12 Trapezium
13 Carpometacarpal joint of thumb



Thenar and hypothenar eminences

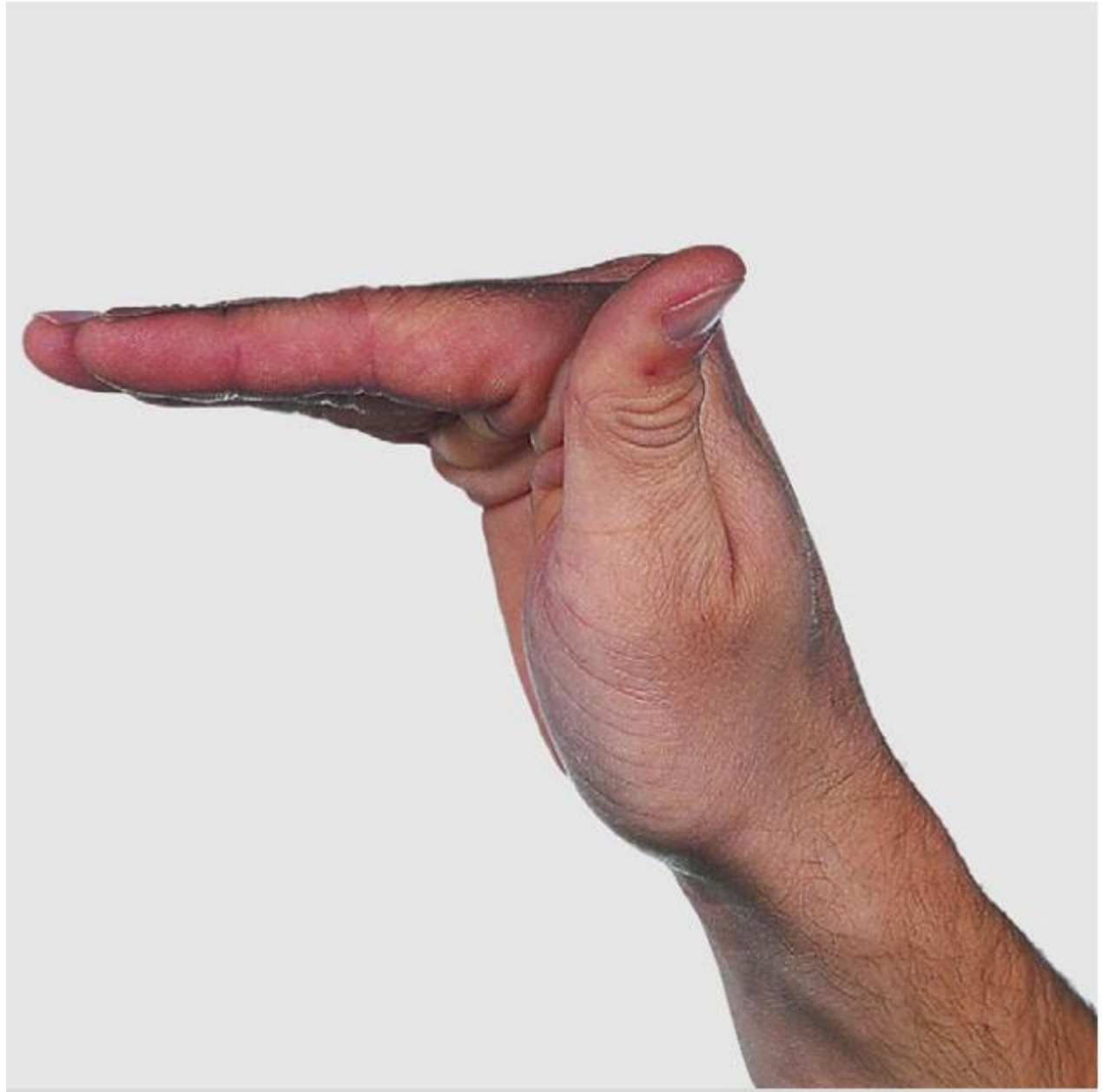
1 Abductor pollicis brevis
2 Flexor pollicis brevis
3 Opponens pollicis
4 Adductor pollicis oblique head
5 Adductor pollicis transverse head
6 Abductor digiti minimi
7 Flexor digiti minimi
8 Opponens digiti minimi
9 Flexor retinaculum
10 Palmar aponeurosis
11 Flexor fibrous sheaths



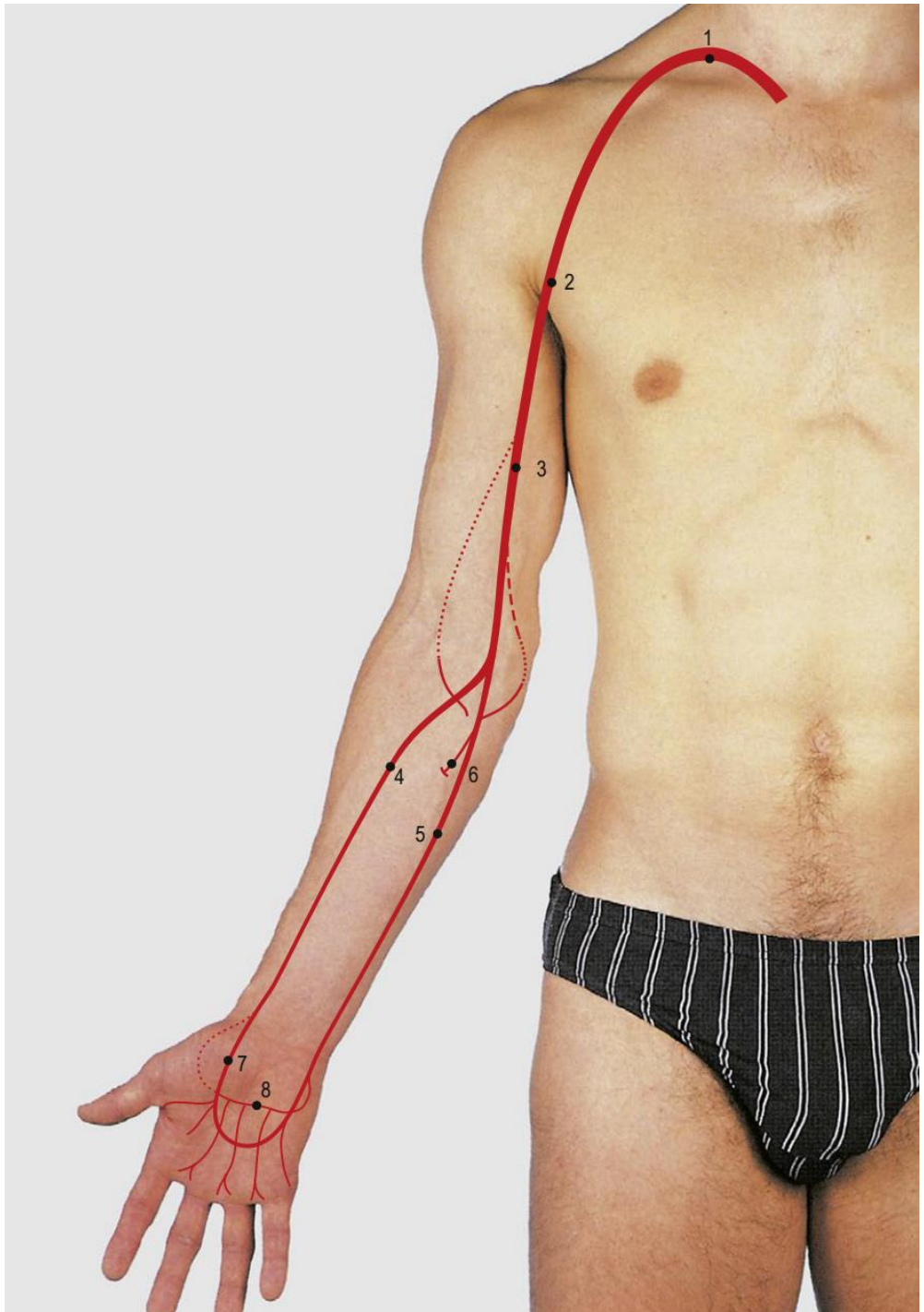
Palmar and dorsal interossei







Action of the lumbrical muscles



Blood Supply of the Upper arm

1 Subclavian
2 Axillary
3 Brachial
4 Radial
5 Ulnar
6 Common interosseous
7 Superficial palmar arch
8 Deep palmar arch


C3
C4
T3
C5
T2
T1
C6
C8
C7


C4
T3
C5
T2
T1
C6
C8
C7

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